



MathJax basic tutorial and quick reference

[+4182] [39] MJD

[2012-08-28 01:46:08]

[support faq mathjax reference]

[<https://math.meta.stackexchange.com/questions/5020/mathjax-basic-tutorial-and-quick-reference>]

(Deutsch: [MathJax: LaTeX Basic Tutorial und Referenz](#) ^[1])

To see how any formula was written in any question or answer, including this one, right-click on the expression and choose "Show Math As > TeX Commands". (When you do this, the '\$' will not display. Make sure you add these: see the next point. There are also [other ways](#) ^[2] to view the code for the formula or the whole post.)

To try formatting, visit the [formatting sandbox](#) ^[3] post, select one of the answers that says “free for editing” and use the “edit” button to edit the answer however you like. Don't forget to change it back when you are finished, so it can be used again.

1. For inline formulas, enclose the formula in $\$...\$$. For displayed formulas, use $\$...\$$.

- These render differently. For example, type the following to show *inline* mode:

$\$ \sum_{i=0}^n i^2 = \frac{(n^2+n)(2n+1)}{6} \$$

$$\sum_{i=0}^n i^2 = \frac{(n^2+n)(2n+1)}{6}$$

- or type the following for display mode:

$\$ \$ \sum_{i=0}^n i^2 = \frac{(n^2+n)(2n+1)}{6} \$ \$$

$$\sum_{i=0}^n i^2 = \frac{(n^2 + n)(2n + 1)}{6}$$

2. For Greek letters, use `\alpha`, `\beta`, ..., `\omega`: α , β , ..., ω .

- For uppercase letters, use `\Gamma`, `\Delta`, ..., `\Omega`.
- Other Greek capital letters are the same as the Latin ones: A, B, E, Z and so on: $A, B, E, Z, \dots$
- Some Greek letters have variant forms: `\epsilon` `\varepsilon` ϵ , ε , `\phi` `\varphi` ϕ , φ , and others.

3. For superscripts and subscripts, use `^` and `_`. For example, x_i^2 , $\log_2 x$: $\log_2 x$. For the prime symbol, use an apostrophe x' x'' x''' : x' x'' x''' .

4. Groups. Superscripts, subscripts, and other operations apply only to the next “group”. A “group” is either a single symbol, or any formula surrounded by curly braces $\{...\}$.

- If you do 10^{10} , you will get a surprise: 10^{10} . But $10^{\{10\}}$ gives what you probably wanted: 10^{10} .
- Use curly braces to delimit a formula to which a superscript or subscript applies: $x^y z$ is an error; $\{x^y\}^z$ is x^{yz} , and $x^{\{y^z\}}$ is x^{y^z} . Observe the differences between x_i^2 , $x_{\{i^2\}}$ x_{i^2} and $\{x_i\}^2$ x_i^2 .

5. Parentheses Ordinary symbols `( ) [ ]` make parentheses and brackets $(2 + 3)[4 + 4]$. Use `\{` and `\}` for curly braces $\{ \}$.

- These do *not* scale with the formula in between, so if you write $\left(\frac{\sqrt{x}}{y^3}\right)$ the parentheses will be too small: $\left(\frac{\sqrt{x}}{y^3}\right)$. Using $\left(\frac{\sqrt{x}}{y^3}\right)$ will make the sizes adjust automatically to the formula they enclose: $\left(\frac{\sqrt{x}}{y^3}\right)$ is $\left(\frac{\sqrt{x}}{y^3}\right)$.
- $\left$ and $\right$ apply to all the following sorts of parentheses: $($ and $)$ (x) , $[$ and $]$ $[x]$, $\{$ and $\}$ $\{x\}$, $|$ $|x|$, $\|$ $\|x\|$, $\langle$ and $\rangle$ $\langle x \rangle$, $\lceil$ and $\rceil$ $\lceil x \rceil$, and $\lfloor$ and $\rfloor$ $\lfloor x \rfloor$. $\middle$ can be used to add additional dividers. There are also invisible parentheses, denoted by $\cdot$: use $\left.x^2\right.\rightvert_3^5 = 5^2 - 3^2$ to get

$$x^2\Big|_3^5 = 5^2 - 3^2$$

6. **Sums and integrals** $\sum$ and $\int$; the subscript is the lower limit and the superscript is the upper limit, so for example $\sum_{i=1}^n$. Don't forget $\{...\}$ if the limits are more than a single symbol. For example, $\sum_{i=0}^{\infty} i^2$ is $\sum_{i=0}^{\infty} i^2$.

- Similarly, $\prod$, $\int$, $\bigcup$, $\bigcap$, $\iint$, $\iiint$, $\dots \int$.

7. **Fractions** There are three ways to make fractions ^[4]. $\frac{a}{b}$ applies to the next two groups, and produces $\frac{a}{b}$; for more complicated numerators and denominators use $\{...\}$: $\frac{a+1}{b+1}$ is $\frac{a+1}{b+1}$.

- If the numerator and denominator are complicated, you may prefer $\over$, which splits up the group that it is in: $\frac{a+1}{b+1}$ is $\frac{a+1}{b+1}$.
- For continued fractions, use $\cfrac$ instead of $\frac$ ^[5].

8. Fonts

- Use $\mathbb{}$ or $\mathbf{}$ for "blackboard bold": $\mathbb{C}\mathbb{H}\mathbb{N}\mathbb{Q}\mathbb{R}\mathbb{Z}$.
- Use $\mathbf{}$ for boldface: $\mathbf{C}\mathbf{H}\mathbf{N}\mathbf{Q}\mathbf{R}\mathbf{Z}\mathbf{c}\mathbf{h}\mathbf{n}\mathbf{q}\mathbf{r}\mathbf{z}$.
 - For expression based characters, use $\boldsymbol$ instead: $\boldsymbol{\alpha}$
- Use $\mathit{}$ for italics: $\mathit{C}\mathit{H}\mathit{N}\mathit{Q}\mathit{R}\mathit{Z}\mathit{c}\mathit{h}\mathit{n}\mathit{q}\mathit{r}\mathit{z}$.
- Use $\mathbf{}$ for boldfaced italics: $\mathbf{C}\mathbf{H}\mathbf{N}\mathbf{Q}\mathbf{R}\mathbf{Z}\mathbf{c}\mathbf{h}\mathbf{n}\mathbf{q}\mathbf{r}\mathbf{z}$.
- Use $\mathtt{}$ for "typewriter" font: $\mathtt{C}\mathtt{H}\mathtt{N}\mathtt{Q}\mathtt{R}\mathtt{Z}\mathtt{c}\mathtt{h}\mathtt{n}\mathtt{q}\mathtt{r}\mathtt{z}$.
- Use $\mathrm{}$ for roman font: $\mathrm{C}\mathrm{H}\mathrm{N}\mathrm{Q}\mathrm{R}\mathrm{Z}\mathrm{c}\mathrm{h}\mathrm{n}\mathrm{q}\mathrm{r}\mathrm{z}$.
- Use $\mathsf{}$ for sans-serif font: $\mathsf{C}\mathsf{H}\mathsf{N}\mathsf{Q}\mathsf{R}\mathsf{Z}\mathsf{c}\mathsf{h}\mathsf{n}\mathsf{q}\mathsf{r}\mathsf{z}$.
- Use $\mathcal{}$ for "calligraphic" letters: $\mathcal{C}\mathcal{H}\mathcal{N}\mathcal{Q}\mathcal{R}\mathcal{Z}$ (Uppercase only).
- Use $\mathscr{}$ for script letters: $\mathscr{C}\mathscr{H}\mathscr{N}\mathscr{Q}\mathscr{R}\mathscr{Z}\mathscr{c}\mathscr{h}\mathscr{n}\mathscr{q}\mathscr{r}\mathscr{z}$.
- Use $\mathfrak{}$ for "Fraktur" (old German style) letters: $\mathfrak{C}\mathfrak{H}\mathfrak{N}\mathfrak{Q}\mathfrak{R}\mathfrak{Z}\mathfrak{c}\mathfrak{h}\mathfrak{n}\mathfrak{q}\mathfrak{r}\mathfrak{z}$.

9. **Radical signs / roots** Use $\sqrt{}$, which adjusts to the size of its argument: $\sqrt{x^3}$; $\sqrt[3]{\frac{x}{y}}$. For complicated expressions, consider using $\sqrt[1/2]{}$ instead.

10. Some **special functions** such as "lim", "sin", "max", "ln", and so on are normally set in roman font instead of italic font. Use $\lim$, $\sin$, etc. to make these: $\sin x \sin x$, not

$\sin x$ *sin* x . Use subscripts to attach a notation to $\lim$: $\lim_{x \rightarrow 0}$

$$\lim_{x \rightarrow 0}$$

Nonstandard function names can be set with $\operatorname{foo}(x)$ $\operatorname{foo}(x)$.

11. There are a very large number of **special symbols and notations**, too many to list here; see the short listing [L^AT_EX and \$\mathcal{A}\mathcal{M}\mathcal{S}\$ - L^AT_EX Symbols](#) ^[6] prepared by Dr. Emre Sermutlu, or the exhaustive listing [The Comprehensive L^AT_EX Symbol List](#) ^[7] by Scott Pakin. Some of the most common include:

- $\lt \gt \leq \geq \neq$ $<, >, \leq, \geq, \neq$. You can use $\not$ to put a slash through almost anything: $\not\lt$ $\not\lt$ but it often looks bad.
- $\times \div \pm \mp \cdot$ $\times, \div, \pm, \mp, \cdot$. $\cdot$ is a centered dot: $x \cdot y$
- $\cup \cap \setminus \subset \supseteq \subseteq \supset \in \notin \emptyset$ $\cup, \cap, \setminus, \subset, \supseteq, \subseteq, \supset, \in, \notin, \emptyset$
- $\binom{n+1}{2k}$ or $\binom{n+1}{2k}$ $\binom{n+1}{2k}$
- $\rightarrow \leftarrow \Rightarrow \Leftarrow \mapsto$ $\rightarrow, \leftarrow, \Rightarrow, \Leftarrow, \mapsto$
- $\land \lor \neg \forall \exists \top \bot \dashv \wedge \vee \neg \forall \exists \top \bot$ $\wedge, \vee, \neg, \forall, \exists, \top, \bot$
- $\star \ast \oplus \circ \bullet$ $\star, \ast, \oplus, \circ, \bullet$
- $\approx \sim \simeq \cong \prec \succ$ $\approx, \sim, \simeq, \cong, \prec, \succ$
- $\infty \aleph_0 \nabla \partial \Im \Re \mathfrak{I} \mathfrak{R}$ $\infty, \aleph_0, \nabla, \partial, \Im, \Re, \mathfrak{I}, \mathfrak{R}$
- For modular equivalence, use $\pmod$ like this: $a \equiv b \pmod n$. For the binary mod operator, use $\bmod$ like this: $a \bmod 17$.
- Use $\dots$ for the triple dots in $a_1, a_2, \dots, a_n$ and $a_1 + a_2 + \dots + a_n$
- Script lowercase l is ℓ .

[Detexify](#) ^[8] lets you draw a symbol on a web page and then lists the $T_{E}X$ symbols that seem to resemble it. These are not guaranteed to work in MathJax, but it's a good place to start. To check that a command is supported, note that MathJax.org maintains a [list of currently supported L^AT_EX commands](#) ^[9], and one can also check Dr. Carol JVF Burns's page of [T_EX Commands Available in MathJax](#) ^[10].

12. **Spaces** MathJax usually decides for itself how to space formulas, using a complex set of rules. Putting extra literal spaces into formulas will not change the amount of space MathJax puts in: $a _ b$ and $a _ _ _ b$ are both ab . To add more space, use $\,$ for a thin space $a \, b$; $\;$ for a wider space $a \; b$. $\quad$ and $\qquad$ are large spaces: $a \quad b, a \qquad b$.

To set plain text, use ... : $\{x \in s \mid x \text{ is extra large}\}$. You can nest $\$... \$$ inside of ... , for example to access spaces.

13. **Accents and diacritical marks** Use `\hat` for a single symbol $\hat{x}$, `\widehat` for a larger formula $\widehat{xy}$. If you make it too wide, it will look silly. Similarly, there are `\bar` $\bar{x}$ and `\overline` $\overline{xyz}$, and `\vec` $\vec{x}$ and `\overrightarrow` $\overrightarrow{xy}$ and `\overleftarrow` $\overleftarrow{xy}$ and `\overleftrightarrow` $\overleftrightarrow{xy}$. For dots, as in $\frac{d}{dx}xx = \dot{x}^2 + x\ddot{x}$, use `\dot` and `\ddot`.
14. Special characters used for MathJax interpreting can be escaped using the `\` character: `\$`, `\{`, `\}`, `\_`, `\#`, `\&`. If you want `\` itself, you should use `\backslash` (symbol) or `\setminus` ([binary operation](#) ^[11]) for `\`, because `\\` is for a new line.

(Tutorial ends here.)

It is important that this note be reasonably short and not suffer from too much bloat. To include more topics, please create short addenda and post them as answers instead of inserting them into this post.

Contents

Alphabetical list of links to MathJax topics, by title:

- [Absolute values and norms](#) ^[12] • [Additional symbolic decorations](#) ^[13] • [Aligning Equations](#) ^[14]
- [Alternative Ways of Writing in LaTeX](#) ^[15] • [Annotations of reasoning](#) ^[16] • [Arbitrary operators](#) ^[17]
- [Arrays](#) ^[18] • [Big braces](#) ^[19] • [Colors](#) ^[20]
- [Commutative diagrams](#) ^[21] • [Continued fractions](#) ^[22] • [Crossing things out](#) ^[23]
- [Definitions by cases \(piecewise functions\)](#) ^[24] • [Degree symbol](#) ^[25] • [Display style](#) ^[26]
- [Equation numbering](#) ^[27] • [Fussy spacing issues](#) ^[28] • [Highlighting expressions](#) ^[29]
- [Left and right arrows](#) ^[30] • [Limits](#) ^[31] • [Linear programming](#) ^[32]
- [Long division](#) ^[33] • [Matrices](#) ^[34] • [Markov Chains](#) ^[35]
- [Mixing code and MathJax formatting on lines](#) ^[36] • [The `\newcommand` function](#) ^[37]
- [Numbering Equations](#) ^[38] • [Overlaying Symbols](#) ^[39] • [Packs of cards](#) ^[40]
- [Symbols](#) ^[41] • [System of equations](#) ^[42] • [Tables](#) ^[43]
- [Tags and references](#) ^[44] • [Tensor indices](#) ^[45] • [Units](#) ^[46]
- [Vertical bars](#) ^[47] • [Vertical spacing](#) ^[48]

[1] <https://www.mathelounge.de/509545/mathjax-latex-basic-tutorial-und-referenz-deutsch>

[2] [/questions/659/how-to-view-latex-source-of-equations](#)

[3] <https://math.meta.stackexchange.com/q/370/25554>

[4] [/questions/12978/should-dfrac-be-edited-in](#)

[5] [/questions/5020/mathjax-basic-tutorial-and-quick-reference/5058#5058](#)

[6] <https://pic.plover.com/MISC/symbols.pdf>

[7] <https://www.ctan.org/tex-archive/info/symbols/comprehensive/symbols-a4.pdf>

[8] <https://detexify.kirelabs.org/classify.html>

[9] <https://docs.mathjax.org/en/latest/input/tex/macros/index.html>

[10] <https://www.onemathematicalcat.org/MathJaxDocumentation/TeXSyntax.htm>

[11] <https://tex.stackexchange.com/questions/511328/difference-between-commands-setminus-and-backslash/511332#511332>

[12] [/questions/5020/mathjax-basic-tutorial-and-quick-reference/15078#15078](#)

[13] [/questions/5020/mathjax-basic-tutorial-and-quick-reference/13081#13081](#)

[14] [/questions/5020/mathjax-basic-tutorial-and-quick-reference/5024#5024](#)

[15] [/questions/5020/mathjax-basic-tutorial-and-quick-reference/27910#27910](#)

[16] /questions/5020/mathjax-basic-tutorial-and-quick-reference/21258#21258
 [17] /questions/5020/mathjax-basic-tutorial-and-quick-reference/15077#15077
 [18] /questions/5020/mathjax-basic-tutorial-and-quick-reference/5044#5044
 [19] /questions/5020/mathjax-basic-tutorial-and-quick-reference/11423#11423
 [20] /questions/5020/mathjax-basic-tutorial-and-quick-reference/10116#10116
 [21] /questions/5020/mathjax-basic-tutorial-and-quick-reference/16888#16888
 [22] /questions/5020/mathjax-basic-tutorial-and-quick-reference/5058#5058
 [23] /questions/5020/mathjax-basic-tutorial-and-quick-reference/13183#13183
 [24] /questions/5020/mathjax-basic-tutorial-and-quick-reference/5025#5025
 [25] /questions/5020/mathjax-basic-tutorial-and-quick-reference/19678#19678
 [26] /questions/5020/mathjax-basic-tutorial-and-quick-reference/25054#25054
 [27] /questions/5020/mathjax-basic-tutorial-and-quick-reference/27793#27793
 [28] /questions/5020/mathjax-basic-tutorial-and-quick-reference/5057#5057
 [29] /questions/5020/mathjax-basic-tutorial-and-quick-reference/22395#22395
 [30] /questions/5020/mathjax-basic-tutorial-and-quick-reference/13310#13310
 [31] /questions/5020/mathjax-basic-tutorial-and-quick-reference/12850#12850
 [32] /questions/5020/mathjax-basic-tutorial-and-quick-reference/27756#27756
 [33] /questions/5020/mathjax-basic-tutorial-and-quick-reference/21096#21096
 [34] /questions/5020/mathjax-basic-tutorial-and-quick-reference/5023#5023
 [35] /questions/5020/mathjax-basic-tutorial-and-quick-reference/31141#31141
 [36] /questions/5020/mathjax-basic-tutorial-and-quick-reference/25251#25251
 [37] /questions/5020/mathjax-basic-tutorial-and-quick-reference/11638#11638
 [38] /questions/5020/mathjax-basic-tutorial-and-quick-reference/11491#11491
 [39] /questions/5020/mathjax-basic-tutorial-and-quick-reference/32210#32210
 [40] /questions/5020/mathjax-basic-tutorial-and-quick-reference/22516#22516
 [41] /questions/5020/mathjax-basic-tutorial-and-quick-reference/11284#11284
 [42] /questions/5020/mathjax-basic-tutorial-and-quick-reference/6267#6267
 [43] /questions/5020/mathjax-basic-tutorial-and-quick-reference/29979#29979
 [44] /questions/5020/mathjax-basic-tutorial-and-quick-reference/11491#11491
 [45] /questions/5020/mathjax-basic-tutorial-and-quick-reference/30661#30661
 [46] /questions/5020/mathjax-basic-tutorial-and-quick-reference/27212#27212
 [47] /questions/5020/mathjax-basic-tutorial-and-quick-reference/35103#35103
 [48] /questions/5020/mathjax-basic-tutorial-and-quick-reference/25048#25048

(41) Some capital Greek letters are the same as the Roman equivalents, so they are not separated in $LATEX$. For a capital beta, one must use something like `\mathrm{B}`: **B - robjohn**

(11) Two related questions: [How do I insert a table when asking a question?](#) and [How to show the integral symbol on this site?](#) - **Martin Sleziak**

(42) A quick addition to point 11: If you want to use a sin-like symbol that is not already defined, the command is `\operatorname`: e.g., `\operatorname{Spec}` A gives $\operatorname{Spec} A$. - **Charles Staats**

(2) @CharlesStaats I considered putting that in, but I wasn't sure it met the threshold for frequently-usedness. Suppose someone wanted it, but didn't know about it. Then in the "worst" case their operator would appear in italics, which did not seem to me to be a serious problem, or one that would impede the person from completing their post. - **MJD**

(1) I can't speak for others, but I use it all the time (far more often than negative spaces, for instance), and found it extremely annoying to want it before I knew the command. Another point is that this is much more useful in MathJax than in Latex (where one would typically define the new operator in the preamble), so it may be hard to find when looking up resources for latex commands. - **Charles Staats**

(24) It might be useful to mention hanging subscripts for things like ${}_5C_3$. You could also mention `\frac` vs `\dfrac`. - **axblount**

(2) @axblount I wonder if a more generally useful thing would be to point out that empty groups `{ }` count as formulas and can appear anywhere that any other formula can. I think `\dfrac` is a refinement that doesn't need to be mentioned in the main post. - **MJD**

(3) @MJD I agree on both. I guess it's best to avoid the technicalities of typesetting or this will quickly cease to be a "basic tutorial." - **axblount**

(8) My basic idea is that if a beginner can express a formula clearly, then someone else can come in and clean up the typesetting afterwards. I am considering getting rid of the section about `\big`, `\left`, and `\right` for this reason, and

trimming the section on spacing. - **MJD**

(10) Most of the references to TeX or LaTeX in this and the answers ought to be to MathJax (the exception that I can see being the output of Detexify). I know this is a bit pedantic, but would it be alright to correct this? - **Andrew Stacey**

(6) @AndrewStacey Thanks for pointing this out. Let's by all means be as correct as possible, particularly when there's no extra cost. - **MJD**

(4) @MJD Okay, I've had a go (also the answer about arrays). I wonder also whether or not it is worth a sentence at the end pointing out that whilst MathJax does its best to emulate TeX, it isn't TeX and so while knowing how something is done in TeX gives you a starting point, it isn't a guarantee that the same thing works in MathJax. (As a case in point, questions about MathJax are generally *off-topic* over on TeX-SX.) - **Andrew Stacey**

(7) @AndrewStacey I wouldn't. They are close enough that it seems to me to be a needless refinement. I might even argue that MathJax is $T_{\text{E}}X$, although an alternative implementation. We're willing to accept that other programming languages (JavaScript, for example) that have slightly incompatible implementations are nevertheless the same language; why not in this case as well? - **MJD**

(11) @MJD Except that this is meant as a tutorial for those who aren't familiar with the distinction (and there really is a distinction: "slightly incompatible implementations" doesn't really fit the bill here). One thing tutorials often include is a "Where to find out more" section. This doesn't. Someone who doesn't know the distinction might be tempted to search for help on TeX or LaTeX instead and wonder why it doesn't work. - **Andrew Stacey**

(7) @AndrewStacey All the tips given here would work in any $T_{\text{E}}X/L_{\text{A}}T_{\text{E}}X$ environment with the proper packages. MathJax is just the service used to render it. You wouldn't say "MikTeX tutorial" or "texlive tutorial". - **axblount**

(6) @axblount But that's precisely the wrong way around to think about it! The likelihood is that someone will look at this tutorial to figure out how to write something on the Maths-SX site: i.e., to use MathJax. If they can't find help here, where do they go? If they have the idea that MathJax is "just a javascript implementation of TeX" then they might think to look for help with TeX, but that is quite possibly *not* going to be helpful. - **Andrew Stacey**

(2) @AndrewStacey Honest question because I have no idea: What sorts of LaTeX math expressions is MathJax incapable of rendering? - **axblount**

(9) @axblount For a start, you've changed the goalposts: "LaTeX **math** expressions". LaTeX is so much more than just a way of typesetting maths! Second, I don't really know but it wouldn't take me long to cook one up. I don't use MathJax so I haven't explored it. But I know, for example, that it can't handle catcode changes. Which means that I can't make (and) *automatically* resizeable. I can in LaTeX. - **Andrew Stacey**

(3) @axblount Anything that requires loading an additional package which MathJax doesn't load (either by default, on our configuration, or impossible to load b/c an implementation hasn't been coded yet). Some random examples include those of stmaryrd like `\inplus`, `\boxast`, `\owedge`. - **Willie Wong**

(3) A not-just-random-symbols example of something that works in LaTeX but not in MathJax: LaTeX has the `empheq` package ([Link](#)) whose functions are, best as I can tell, not replicated in MathJax. And just for fun I give a MathJax expression that won't render in LaTeX: try `$$ \begin{align} \dots \end{align} $$`. - **Willie Wong**

(2) Shoot, the "click the edit link" advice is not very good because users don't get an edit link until they have some reputation. It's unfortunate that new users don't even get a "view source" link. I wonder if that advice should be demoted? - **MJD**

(2) @WillieWong That's cheating. The `alignat` environment is for when aligning stuff when you are *already* in a mathematical environment and would have been a better choice for MathJax to use here. But it's an important example of the limitations of MathJax: it is *only* for mathematical formatting. - **Andrew Stacey**

(3) I don't *really* mind `\over` being mentioned in item 8, but I *had to* replace it with `\frac` in the first two examples of fractions. [What is the difference between `\over` and `\frac`?](#) // By the way, I think this post should be tagged (`faq`) but it's stuck on (`faq-proposed`). - **user53153**

(3) @5PM Okay, but why do you care so much, and why should we care that you care so much? Is there a reason to prefer `\frac` to `\over`? - **MJD**

(3) @MJD The TeX.SE thread to which I linked gives some reasons for preferring `\frac`. The simplest reason is to maintain consistent format of LaTeX commands, which is the prefix form `\command{argument}{maybe more arguments}`. For example, the LaTeX way is `\textit{something}` while the old TeX way was `{\it something}`. Same with `\frac` and `\over`. // Someone who is used to TeX in one or another way can happily proceed without worrying about what I or anyone else thinks. But someone who's just beginning to learn may well begin with the modern approach. - **user53153**

(3) All those reasons seem to apply to $T_{\text{E}}X/L_{\text{A}}T_{\text{E}}X$, but not to MathJax, and not to anything we would likely be doing in MathJax or on this site. For example, the fact that the existence of `\over` complicates the design of macros is completely irrelevant to us here. - **MJD**

(90) I wish I saw this post when I first joined. This post should be a main link on the home page. There should be a button under each box: NEW TO LATEX, CLICK HERE FOR EXAMPLES. This is extremely useful, concise. - **user1527227**

(3) @robjohn I never use `\mathrm`. I can see the value of mentioning `\text`, because beginners often write things like

$\{x|xishyperellipticallyconstrained\}$, but `\mathrm` doesn't help here, and I'm not sure why it is a good use of space in the tutorial. Is it important for some reason I don't appreciate? - **MJD**

(16) @MJD: I use `\mathrm` in many places; e.g. dx in integrals and derivatives and for operator names that don't need the full force of `\operatorname`. `\mathrm` was intended for roman symbols in math mode; `\text` was intended for text because of the way it spaces things. See [this TEX thread](#). Since I don't believe we can use preambles in MathJax, we can't use `\DeclareMathOperator`, though we can use `\newcommand`, but that is orthogonal to the use of `\mathrm` vs `\text` for math symbols. - **robjohn**

(1) Another useful resource is Wikipedia's [Help:Displaying a formula](#). - **r.e.s.**

(2) Might be late, but when I search "mathjax tutorial" on Google, this meta post is the first result returned. Congrats! - **Andrew T.**

I think this thread: meta.math.stackexchange.com/questions/20896/... should be linked here. - **Surb**

(2) Wow, this is the first post on main and meta together to reach more than 1k votes. - **wythagoras**

Why isn't this $\therefore$ up there. therefore should be a must. - **user312097**

Because it's not important. Someone who doesn't know how to write $\therefore$ will have no trouble thinking of how to write "therefore" instead. - **MJD**

How to write H is normal in G ? I'm talking about the inverted delta sign. Thanks. - **null**

@Sayantani It's $\lhd$. I've added this to section 12. - **MJD**

How do you add a "is proportional to" `\propto` in a Markdown document? - **M. Beausoleil**

@user1527227 We kinda did it! math.stackexchange.com/help/notation - **Simply Beautiful Art**

$\left(\frac{\sqrt{x}}{y}\right)$ - **user167920**

For everyone new to Latex, I have recently written a [realtime latex editor](#) where you can quickly create your math notations and formulas. Enjoy! - **Avatar**

What is the purpose of the first paragraph?? - **Peter Mortensen**

@peter To enable a user of the web site to continue learning on their own from examples that they see on every other post here. - **MJD**

(1) Is there a comic sans font? I believe the addition of this could have very positive effects on math.stackexchange.com - **Flinn Bella**

this was incredibly help to me after getting nowhere with some LaTeX and Tex Stack Exchange tutorials, online books and other resources! The only thing I was looking for that this didn't cover, and I did see mentioned somewhere in a deep LaTeX reference book was how to make equations bigger; when you have factions with power functions and subscripts it can all end up rather small on SE Mathematics using inline i.e single '\$' notation at least. - **wide_eyed_pupil**

I have finded a very nice site where there are a list of complete command to write in MathJax. Very interesting and very fast: the title is TeX Commands available in MathJax. onemathematicalcat.org/MathJaxDocumentation/TeXSyntax.htm - **Sebastiano**

(1) I noticed that quite a few people on main are using `\backslash` instead of `\setminus` for, well, the operation of setminus. The spacing is different: compare $A \backslash B$ with $A \setminus B$, and `\setminus` is designed for this use. I edited the tutorial to mention this, and also added a link to a [TeX.SE question](#) about this. - **user1729**

(1) Is it just me, or does the sentence "There are also other possibilities how to view the code for the formula or the whole post." read badly? I feel that something like "...possibilities for how.." would read better. - **user1729**

Related: [Diagonal lines in table cell on mathjax](#) - [Mathematics Meta Stack Exchange](#) - **user202729**

Perhaps $\uparrow$ and $\downarrow$ could be added too in the section of arrows - **Verónica R. M.**

Here, we can more [express mathematical expression](#) - **code_conundrum**

@MJD Would you mind sharing your reasoning behind revision #102 (i.e. why you rolled back the addition of `\because`)? - **iBug**

Main reason: nobody ever uses it. Secondary reason: nobody needs it; it's an obscure and inferior replacement for "because". Tertiary reason: we already have links to lists of TeX symbols that include it. - **MJD**

@AndrewStacey that very same thing happened to me yesterday and then i found this tutorial which was a godsend... "Someone who doesn't know the distinction might be tempted to search for help on TeX or LaTeX instead and wonder why it doesn't work." - **Jules Manson**

`\nsim` for not similar. - **Lori**

(1) @DanBoschen I do not agree with adding $\circledast$. This page is not meant to be exhaustive, and I don't think $\circledast$ is described by "Some of the most common". And there are other symbols `\odot`, `\otimes`, $\odot$, $\otimes$ that are more common in my experience. - **Calvin Khor**

@CalvinKhor It is commonly used to denote the circular convolution process, but if there isn't room to include it I understand. - **Dan Boschen**

(1) @DanBoschen I have never seen that in any math paper or book (but I am young..). [Wikipedia's entry on the circular convolution](#) does not have it; it instead uses $*$ which I would say is commonly used. I don't feel strongly enough to edit, so I have only voiced my disagreement. - **Calvin Khor**

@CalvinKhor Here's a very reputable source : ccrma.stanford.edu/~jos/st/Convolution.html (it's common in signal processing applications) - **Dan Boschen**

OK. In [this ScienceDirect link](#) there are many competing notations: $*$, $\otimes_L$, $\circledast$; Wikipedia in [Convolution](#) uses $*_N$ to distinguish from the usual convolution. Google did show me [someone on dsp.SE](#) using $\circledast$. [Someone on TeX.SE](#) also asked in 2017 where it is used. - **Calvin Khor**

(2) @iBug why would you want to teach a newbie to use `\ldots`, and in any case in your example it renders the same as `\dots`? I can see `\lhd` being used but why put it with the euivalence relations? (similar question for `\prec` but IDK when that was added) (And while I'm at it a better edit description would be nice. Even the default one beats your apple) - **Calvin Khor**

@CalvinKhor There are a lot of symbols added "for completeness". In fact, my edit added `\dots` (where there used to be only `\ldots`). Same for `\lhd` which was already there and I only added `\rhd` and an explanation. - **iBug**

(1) Ah. Sorry for misreading the edit. It would IMO be better to remove mention of `\ldots` and `\cdots` - **Calvin Khor**

@calvin I agree, including more than one of the dots macros is unnecessary and at least one should be removed. - **MJD**

(1) @iBug "completeness" is not a goal here. It is an anti-goal. The goal is to provide as *little* as possible. - **MJD**

@MJD I understand how "completeness" is not the goal here, but I find it frustrating if I were to learn `\lt` without `\gt` (just for example). Sure I'll remove `\ldots` but I'd keep `\rhd` though. Same reasoning when I previously added `\gets` and `\impliedby`. - **iBug**

@iBug What is the difference between `\ldots` and `\dots`? - **Tyma Gaidash**

(1) `\lhd` was in there because it is a very frequently-used symbol in group theory. As far as I know `\rhd` is not common. And your imagined frustration is not a reliable guide to what should be in the tutorial. It is important that this article be reasonably short and not suffer from too much bloat. - **MJD**

@TymaGaidash You type one less letter, and otherwise identical. - **iBug**

(2) @TymaGaidash iBug is wrong, both in MathJax and LaTeX. The difference is that `\dots` tries to intelligently choose between `\ldots` (l for lower) and `\cdots` (c for center) depending on the surrounding text e.g. $a + \ldots + ba + \dots + b$ vs $a + \cdots + ba + \cdots + b$. - **Calvin Khor**

(1) @CalvinKhor That's great, thanks, because it means we can get rid of `\cdots`. - **MJD**

@CalvinKhor My original idea was not to divide up the relations by equivalence / nonequivalence, but to put the elementary arithmetic relations $<, >, \leq, \geq, \neq$ in one place, and the more advanced symbols $\sim, \equiv, \cong, \prec, \triangleleft$ in another. My sense was that $\succ$ is much less frequently used, and that someone looking for it would be a more advanced user who could guess the name, so I left it out. Over time the first section was bloated with unnecessary additions like `\leqslant` and `\leqq`. - **MJD**

@MJD Why remove the "if manual adjustments are required" parenthesis section; it was there for some time? - **Tyma Gaidash**

It has been in there since day 1 in fact. But I don't think it has ever been useful. `\left` and `\right` are good enough for all possible Math SE uses. I think there's no unclear formula anyone could write that would become clear by using `\biggl`. - **MJD**

@MJD, why the roll-back to revert my edit? Learning how to add a stretched vertical bar to indicate the boundary values for a definite integral took a lot of work and was hard to find. - **Gabriel Staples**

(1) It is important that this note be reasonably short and not suffer from too much bloat. I support your addendum post. - **MJD**

For biconditional statements, as well as $\iff$ there is also $\leftrightarrow$ `\leftrightharpoonrightarrow` for a tidier look. - **Evan Schwartzentruber**

There is also $\Leftrightarrow$ `\Leftrightarrow` - **Tyma Gaidash**

(3) There is also [Short and helpful advice on using MathJax on the site](#). - **Peter Mortensen**

Even though others may exist outside of SE, why not make an HTML, commands like `</>`, guide on MSE? - **Tyma Gaidash**

Suggestions: Replace `\neq` with `\ne` and the `\mathscr` on lowercase letters does not work - **Tyma Gaidash**

Added additional symbols and info to paragraph 6. to construct a big operator. - **zwim**

I think that would be better in a separate post. It is important that the main post be reasonably short and not suffer from too much bloat - **MJD**

[+467] [2012-08-28 04:17:33] MJD

Matrices

1. Use `$$\begin{matrix}...\end{matrix}$$` In between the `\begin` and `\end`, put the matrix elements. End each matrix row with `\\`, and separate matrix elements with `&`. For example,

```

$$
\begin{matrix}
1 & x & x^2 \\
1 & y & y^2 \\
1 & z & z^2
\end{matrix}
$$

```

produces:

$$\begin{matrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{matrix}$$

MathJax will adjust the sizes of the rows and columns so that everything fits.

2. To add brackets, either use `\left...\right` as in section 6 of the tutorial, or replace `matrix` with `pmatrix` $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, `bmatrix` $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, `Bmatrix` $\begin{Bmatrix} 1 & 2 \\ 3 & 4 \end{Bmatrix}$, `vmatrix` $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$, `Vmatrix` $\begin{Vmatrix} 1 & 2 \\ 3 & 4 \end{Vmatrix}$.

3. Use `\cdots` $\cdots$, `\ddots` $\ddots$, `\vdots` $\vdots$ when you want to omit some of the entries:

$$\begin{pmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^n \\ 1 & a_2 & a_2^2 & \cdots & a_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_m & a_m^2 & \cdots & a_m^n \end{pmatrix}$$

4. For horizontally "augmented" matrices, put parentheses or brackets around a suitably-formatted table; see [arrays](#)^[1] below for details. Here is an example:

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 4 & 5 & 6 \end{array} \right]$$

is produced by:

```

$$ \left[
\begin{array}{cc|c}
1&2&3\\
4&5&6
\end{array}
\right] $$

```

The `cc|c` is the crucial part here; it says that there are three centered columns with a vertical bar between the second and third.

5. For vertically "augmented" matrices, use `\hline`. For example

$$\left(\begin{array}{cc} a & b \\ c & d \\ \hline 1 & 0 \\ 0 & 1 \end{array}\right)$$

is produced by

```


$$\begin{pmatrix} a & b \\ c & d \\ \hline 1 & 0 \\ 0 & 1 \end{pmatrix}$$


```

6. For small inline matrices use `\bigl(\begin{smallmatrix} ... \end{smallmatrix}\bigr)`, e.g. $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is produced by:

```
 $\bigl(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\bigr)$ 
```

[1] <http://meta.math.stackexchange.com/a/5044/>

(28) This says "End each matrix row with `\`". But there is no reason to end the LAST row of the matrix that way. The double backslash means: now go on to the next row. But there isn't any next row after the last one. - **Michael Hardy**

(6) I can't edit, but that could be phrased "Separate matrix rows with `\`". - **trichoplax is on Codidact now**
as in section 6 of the tutorial... Which tutorial? Is there a link to this tutorial section? - **Tom Hale**
@tom The tutorial is at the top of this page. It has numbered sections. - **MJD**

(8) @MichaelHardy but a `\` on every line is harmless, and it makes the editing of matrices easier because swapping with the last line can be done with one quick keystroke in many editors. - **Reb.Cabin**

Is it possible to get `smallmatrix` or something? - **linear_combinatori_probabi**

There's something strange about the second matrix (right after "produces:"); it seems to render properly as a matrix on this page, but on the revision permalink math.meta.stackexchange.com/revisions/5023/7 it shows up as raw LaTeX in a code block. - **j.c.**

`vmatrix` does not show the vertical bar. - **alhelal**

`vmatrix` does not show the vertical bar. (

$$\begin{vmatrix} 0 & 3 & 2x+7 \\ 2 & 7x & 9+5x \\ 0 & 0 & 2x+5 \end{vmatrix}$$

=0) হলে x এর মান- - **alhelal**

I am using `<script src="https://polyfill.io/v3/polyfill.min.js?features=es6"></script> <script type="text/javascript" id="MathJax-script" async src="https://cdn.jsdelivr.net/npm/mathjax@3/es5/tex-ctml.js"> </script>` - **alhelal**

Is there a way to combine the horizontal and vertical lines in a single matrix to show the partitions of a matrix? - **Hosein Rahnama**

for a reflected version of `\ddots` that run from SW to NE [this answer](#) on mathematics meta works:
`\newcommand\iddots{\mathinner{\kern1mu\raise1pt{.}\kern2mu\raise4pt{.}\kern2mu\raise7pt{\Rule{0pt}{7pt}{0pt}}\kern1mu}}` - **Manfred Weis**

1

[+344] [2013-10-14 20:15:28] user93957

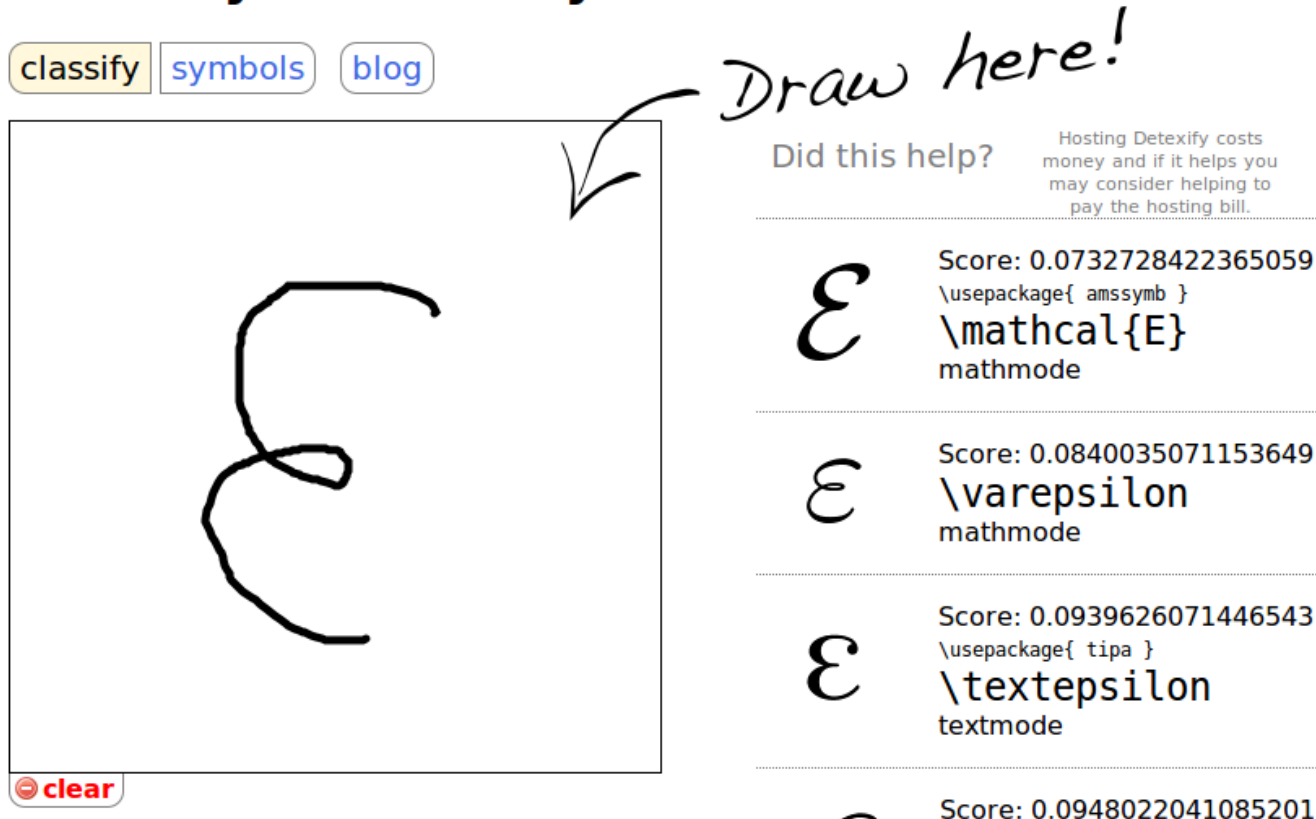
Symbols

In general, you have to search in long tables about a specific symbol you're looking for, things like Ψ , δ , ζ ,

$\geq$, $\subseteq$... And it turns out that this operation can be frustrating and time consuming, which can cause the buddy to abandon writing the complete *L^AT_EX* sentence in his answer, or in some cases, the complete answer itself.

That's why the tool that I will present you in this post was conceived. Basically, it is a *L^AT_EX* handwritten symbol recognition. Example in image:

Detexify² - LaTeX symbol classifier



classify symbols blog

Draw here!

Did this help? Hosting Detexify costs money and if it helps you may consider helping to pay the hosting bill.

$\mathcal{E}$	Score: 0.0732728422365059 <code>\usepackage{ amssymb }</code> <code>\mathcal{E}</code> mathmode
ε	Score: 0.0840035071153649 <code>\varepsilon</code> mathmode
epsilon	Score: 0.0939626071446543 <code>\usepackage{ tipa }</code> <code>\text{epsilon}</code> textmode
ϵ	Score: 0.0948022041085201

clear

Here is the website: [Detexify²](http://detexify.kirelabs.org/classify.html) ^[1] No more frustration.

[1] <http://detexify.kirelabs.org/classify.html>

(7) How to implement usepackage ? I'd like to have \iddots from package mathdots available. - **Gottfried Helms**

(1) @GottfriedHelms see [this question](#) - I think the answer is "you can't do that"... - **Floris**

@Floris: thanks, that is indeed the informative answer! - **Gottfried Helms**

(1) It recognized my horrible drawing with a finger on my notebook's mousepad! Unfortunately the symbol it recognized (mapsfrom) isn't part of MathJax - **Manfred Weis**

2

[+320] [2012-08-28 04:28:50] MJD

Aligned equations

Often people want a series of equations where the equals signs are aligned. To get this, use `\begin{align}...\end{align}`. Each line should end with `\\`, and should contain an ampersand at the point to align at, typically immediately before the equals sign.

For example,

$$\begin{aligned}
 \sqrt{37} &= \sqrt{\frac{73^2 - 1}{12^2}} \\
 &= \sqrt{\frac{73^2}{12^2} \cdot \frac{73^2 - 1}{73^2}} \\
 &= \sqrt{\frac{73^2}{12^2}} \sqrt{\frac{73^2 - 1}{73^2}} \\
 &= \frac{73}{12} \sqrt{1 - \frac{1}{73^2}} \\
 &\approx \frac{73}{12} \left(1 - \frac{1}{2 \cdot 73^2} \right)
 \end{aligned}$$

is produced by

```

\begin{align}
\sqrt{37} &= \sqrt{\frac{73^2-1}{12^2}} \quad \backslash\backslash \\
&= \sqrt{\frac{73^2}{12^2} \cdot \frac{73^2-1}{73^2}} \quad \backslash\backslash \\
&= \sqrt{\frac{73^2}{12^2}} \sqrt{\frac{73^2-1}{73^2}} \quad \backslash\backslash \\
&= \frac{73}{12} \sqrt{1 - \frac{1}{73^2}} \quad \backslash\backslash \\
&\approx \frac{73}{12} \left( 1 - \frac{1}{2 \cdot 73^2} \right) \\
\end{align}

```

The usual \$\$ marks that delimit the display may be omitted here.

(7) The AMS's [Short Math Guide](#) recommends the align environment over eqnarray in LaTeX. In MathJax the spacing seems to be the same, but align requires one less ampersand per line. - **user856**

(3) Thanks. I was not sure whether to discuss that. [A detailed argument against eqnarray is in this article.](#) - **MJD**

Would you mind if I changed your example to use align then? - **user856**

@Rahul: Please go ahead and change anything that seems good to change. This is all CW. - **MJD**

Also, if you think you have a better example, please use it; I used the first one I found. - **MJD**

(7) Correct me if I'm wrong, but I don't believe the \$\$ is necessary before and after the \begin{align}. I've certainly never used it. From experience, the \begin{align} puts you into math-display mode by itself. - **TravisJ**

(7) I sometimes find that one line of this environment is too close to another, making them uncomfortable to read. The interline spacing can be adjusted by using input such as \[1ex] instead of \[. (And of course the 1 can be changed to another value such as 1.5 or .7 in order to get enough space but not too much.) - **David K**

Note - in Jekyll I had to add an additional \ to break lines. - **baxx**

(4) @MJD i put the equation:

```

\begin{align} f(x)&=\left(x^3\right)+\left(x^3+x^2+x^1\right)+\left(x^3+x^2\right)\backslash f'(x)&=\left(\left(3x^2+2x+1\right)\right)+\left(3x^2+2x\right)
\end{align}

```

but the third & is a problem: according to Mathjax, it does not go there. then when i remove it, it says that "a missing close brace or unclosed brace" is present. what am i doing wrong? - **Alexander Day**

(6) @AlexanderDay How did you used that boxed quote? What are commands or formats for it? - **user379641**

@AlexanderDay I notice that right after your second &=, you have two instances of \left(in a row, and one of them is

$$f(x) = (x^3) + (x^3 + x^2 + x^1) + (x^3 + x^2)$$

not closed. Removing one of them, I get: $f'(x) = (3x^2 + 2x + 1) + (3x^2 + 2x)$ Is that what you

$$f''(x) = (6x + 2)$$

expected? - **Dan Henderson**

(4) @AlwaysConfused the box delimits a MathJax formula with a syntax error. - **Dan Henderson**

There is a [difference between aligned and align environment](#). Equation above are just one equation with different representations, rather than multiple aligned equations. I think the correct environment is aligned. - **jdhaio**

(2) The align* variation does two things differently: left-aligns the equations and omits equation numbers; the regular align environment centers the equations and puts an equation number on each line, at least in my version of Jupyter notebooks. - **Reb.Cabin**

@Reb.Cabin I think the thing is that this is MathJaX. When I am writing stuff in LaTeX I always use align*, but I cannot see the difference between align and align* here, in MathJaX world. - **user1729**

Is there a way to do nested alignments? - **Display name**

Often you'll want to add non-aligned text (as an small explanation), is there a command to do so whilst in the align environment? normally i would use \noalign{\text{*text*}} but this doesnt seem to work. - **Vaas**

How can I left align in one line?It centers default. - **上山老人**

update link to [Short Math Guide](#) - **Delfino**

(1) pandoc (or the underlying tex2pdf) actually complains if the \begin{align} is preceded by \$\$ - **TKH**

3

[+267] [2012-08-28 04:34:03] MJD

Definitions by cases (piecewise functions)

Warning: If you make certain kinds of errors while entering code using this environment, you can easily screw-up live update, and your only recourse is to abandon your edit and refresh the page. Clearing out the code and re-entering it will not fix things - you will have to refresh the page. If you are learning how to use this feature it is recommended that you cut-and-paste a working example from here, and modify it bit-by-bit to the text you want.

Use `\begin{cases}...\end{cases}`. End each case with a `\\`, and use `&` before parts that should be aligned.

For example, you get this:

$$f(n) = \begin{cases} n/2, & \text{if } n \text{ is even} \\ 3n + 1, & \text{if } n \text{ is odd} \end{cases}$$

by writing this:

```
f(n) =
\begin{cases}
n/2, & \text{\text{if } $n$ is even} \\
3n+1, & \text{\text{if } $n$ is odd} \\
\end{cases}
```

The brace can be moved to the right:

$$\left. \begin{array}{l} \text{if } n \text{ is even: } \\ \text{if } n \text{ is odd: } \end{array} \right\} \begin{array}{l} n/2 \\ 3n + 1 \end{array} = f(n)$$

by writing this:

```
\left.
\begin{array}{l}
\text{\text{if } $n$ is even:}&n/2\\
\text{\text{if } $n$ is odd:}&3n+1
\end{array}
\right\}
=f(n)
```

To get a larger vertical space between cases we can use `\\[2ex]` instead of `\\`. For example, you get this:

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 3n + 1, & \text{if } n \text{ is odd} \end{cases}$$

by writing this:

```
f(n) =
\begin{cases}
```

```
\frac{n}{2}, & \text{if $n$ is even} \\[2ex]
3n+1, & \text{if $n$ is odd}
\end{cases}
```

(An ‘ex’ is a length equal to the height of the letter x; 2ex here means the space should be two exes high.)

@MJD Do we have to use the additional instruction `\displaystyle` when the formulas displayed are more complex ? - **jibe**

(4) @jibs `\displaystyle` is enabled automatically in displays, for example between `$$...$$`. You should not ever have to use it. - **MJD**

@jibe In general, the separate cases in this notation should be in text style unless they are very very complex (and then, the `{` notation is just wrong anyways). - **yo'**

can this be written with ascii math instead of latex @MJD - **wrufesh**

What an absurd function to use as an example. Nobody would ever consider such a function. - **Robert Frost**

4

[+180] [2012-08-29 18:37:02] axblount

Arrays

It is often easier to read tables formatted in MathJax rather than plain text or a fixed width font. Arrays and tables are created with the `array` environment. Just after `\begin{array}` the format of each column should be listed, use `c` for a center aligned column, `r` for right aligned, `l` for left aligned and a `|` for a vertical line. Just as with matrices, cells are separated with `&` and rows are broken using `\\`. A horizontal line spanning the array can be placed before the current line with `\hline`.

For example,

n	Left	Center	Right
1	0.24	1	125
2	−1	189	−8
3	−20	2000	$1 + 10i$

```
$$
\begin{array}{c|lcr}
n & \text{Left} & \text{Center} & \text{Right} \\
\hline
1 & 0.24 & 1 & 125 \\
2 & -1 & 189 & -8 \\
3 & -20 & 2000 & 1+10i
\end{array}
$$
```

Arrays can be nested to make an array of tables.

For example,

min	0	1	2	3	max	0	1	2	3
0	0	0	0	0	0	0	1	2	3
1	0	1	1	1	1	1	1	2	3
2	0	1	2	2	2	2	2	2	3
3	0	1	2	3	3	3	3	3	3

Δ	0	1	2	3
0	0	1	2	3
1	1	0	1	2
2	2	1	0	1
3	3	2	1	0

As the source for the preceding array is long, please right-click on one of the tables and choose Show Math As ► TeX Commands.

(16) You'll have to wrap the contents of each cell in `\text` if you don't want *allitalics, weird – lookingspacing, an' odd apostrophes*. - **user856**

@RahulNarain: True. I used words just for illustration, but I guess the example was slightly misleading. If you'd like to modify it please go ahead. - **axblount**

(2) Thanks! I like your numeric example better, since the widths of the entries are different enough that the alignment differences are visually clear. - **MJD**

@robjohn how do you use `|` while typing, i don't find it in my keyboard..... - **ABC**

@exploringnet: on my keyboard, it is the shifted backslash. It may be in different places (or absent) depending on your keyboard. On my mobile device (iPhone), it is in the shifted numerics, to the right of the backslash. In mathmode, `\vert` gives `|` and `\mid` gives `|`, but neither works in the column spec for an array. If you cannot type it on your keyboard, you can always copy and paste it from another document. - **robjohn**

It should perhaps be mentioned, that in nested arrays there seems to be no option to synchronize column-widths and/or row-heights over the top-level. I didn't find a solution such that if two arrays are stacked vertically one could make their column-widths matching/fit. - **Gottfried Helms**

(5) This could also be convenient for some people, although it destroys the joy of writing tables in *L^AT_EX* by hand! - **nullgepetto**

@Rahul: why did regulars not press developers to enhance HTML formatting instead of doing inconvenient and resource-devouring detours through MathJax? When a table contains (mostly) formulæ, the use of a formula-formatting engine looks determined. But when one wants *just a table*, why should it run software with completely different purpose? I once tried to speak about it at meta.SE, but was gagged. - **Incnis Mrsi**

@IncnisMrsi What kind of pressure could we apply: bribery, threats, kidnapping? A feature request was made, supported by SE communities, and declined by SE (on technical grounds, as they say). At least we have the MathJax workaround, with all of its flaws: SO and others have nothing. - **user147263**

(9) Center Aligned Table Captions with Left Aligned Contents

Bad	Better
$e^{i\frac{\pi}{2}}$	$e^{\frac{i\pi}{2}}$
$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \, dx$	$\int_{-\pi/2}^{\pi/2} \sin x \, dx$

- **GNUSupporter 8964民主女神 地下教會**

P.S. Table copied from MJD's example below so as to make an example with cells much wider than the caption. - **GNUSupporter 8964民主女神 地下教會**

@MJD Is there any command for a vertical line, like `\hline`? The `{cc|c}` part in the array code is confusing. - **Tyma Gaidash**

There is the `\rm` command instead of the entire code in `\text` - **Tyma Gaidash**

[+170] [2012-08-31 19:16:11] MJD

Fussy spacing issues

These are issues that won't affect the correctness of formulas, but might make them look significantly better or worse. Beginners should feel free to ignore this advice; someone else will correct it for them, or more likely nobody will care.

Don't use `\frac` in exponents or limits of integrals; it looks bad and can be confusing, which is why it is rarely done in professional mathematical typesetting. Write the fraction horizontally, with a slash:

Bad	Better
$e^{i\frac{\pi}{2}}$	$e^{\frac{i\pi}{2}}$
$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \, dx$	$\int_{-\pi/2}^{\pi/2} \sin x \, dx$

The `|` symbol has the wrong spacing when it is used as a divider, for example in set comprehensions. Use `\mid` instead:

Bad	Better
$\{x x^2 \in \mathbb{Z}\}$	$\{x \mid x^2 \in \mathbb{Z}\}$

When using stretchable delimiters (i.e. with `\left` and `\right`), it may be preferable to use `\,``\middle|``\,`. This produces a stretchable vertical bar with a little bit of space around it. Another alternative is to use a colon instead.

Bad	Better
$\left\{ \frac{m}{n} \mid m, n \in \mathbb{Z} \right\}$	$\left\{ \frac{m}{n} \middle m, n \in \mathbb{Z} \right\}$

For double and triple integrals, don't use `\int\int` or `\int\int\int`. Instead use the special forms `\iint` and `\iiint`:

Bad	Better
$\int \int_S f(x) \, dy \, dx$	$\iint_S f(x) \, dy \, dx$
$\int \int \int_V f(x) \, dz \, dy \, dx$	$\iiint_V f(x) \, dz \, dy \, dx$

Use `\,` to insert a thin space before differentials; without this TEX will mash them together:

Bad	Better
$\iiint_V f(x) dz dy dx$	$\iiint_V f(x) \, dz \, dy \, dx$

When using a function (e.g. f , $\sin$, etc) followed by arguments with larger parentheses, insert negative space before the parentheses using `\!`:

Bad	Better
$f\left(\frac{1}{x}\right)$	$f\left(\frac{1}{x}\right)$

When using absolute value, use `\lvert` ... `\rvert` instead of a pair of pipes `|` ... `|`.

Bad	Better
$ \sin x $	$ \sin x $

(5) I think the first adjusted fraction looks better than the original, but I don't like the second. In any case, this minor spacing imbalance is too peripheral to belong in a basic MathJax tutorial IMO. Too likely to scare people away rather than make them feel helped. - **hmakholm left over Monica**

(3) @Henning Do you mean that the fraction example is too unimportant even to appear in an addendum on fussy spacing, or that the fussy spacing article is too unimportant to appear as an addendum to the tutorial? - **MJD**

(2) I was talking specifically about the fraction example. Mostly I'm concerned that somebody will come away thinking, *Eeek! Do I have to worry about THAT to use the site?* But it's also arguable that the disclaimer at the top of the answer ought to take care of that. - **hmakholm left over Monica**

(2) @MJD I like the less space, but what if we want to list the bounds for multiple integrals? Like if we have say 3 integrals and we have 3 separate bounds for each how would we list each one? Or do we have to do `\int_{bound1}^{bound2}\int_{bound3}^{bound4}\int_{bound5}^{bound6}??` - **TheHopefulActuary**

(2) @Kyle I think that's exactly what you do in that case. - **MJD**

(34) Worth nothing you can use `\middle` with `|` to get it to work with `\left` and `\right`, like `\left\{x\middle|\frac{x^2}{2}\in\mathbb{Z}\right\}` - **asmeurer**

(1) Thanks very much! I wanted to do that, but didn't know how. - **MJD**

@asmeurer Don't forget the spacing around the bar. - **user76284**

(2) It seems `\middle \mid` doesn't work. What is the correct way to get the right spacing with automatic vertical resizing? - **asmeurer**

In the case of base e powers I would recommend using $\exp(i\pi/2)$ which is, in my opinion, even better than what's suggested in this post. - **mechanicious**

@asmeurer I always use `\left\{, \dots, \middle|, \dots, \right\}` like in $\left\{x \in \mathbb{R} \middle| \frac{x^2}{2} \in \mathbb{Z}\right\}$. - **Christoph**

I've been using `\;` and `\:` along with `\,,` and I think they have different spacing, so it's pretty versatile but can get you stuck on formatting an answer for ages because you were trying to get the spaces to match on the scale of pixels :| - **sreysus**

6

[+147] [2014-03-23 23:28:03] MJD

Crossing things out

Use `\require{cancel}` in the first formula in your post that requires cancelling; you need it only once per page. Then use:

$$\begin{array}{ll}
 y+\cancel{x} & y + \cancel{x} \\
 \cancel{y+x} & \cancel{y+x} \\
 y+\bcancel{x} & y + \bcancel{x} \\
 y+\xcancel{x} & y + \xcancel{x} \\
 y+\cancelto{0}{x} & y + \cancelto{0}{x}^0 \\
 \frac{1\cancel{9}}{\cancel{9}5} = \frac{15}{5} & \frac{1\cancel{9}}{\cancel{9}5} = \frac{1}{5}
 \end{array}$$

Use `\require{enclose}` for the following:

$$\begin{array}{ll}
 \enclose{horizontalstrike}{x+y} & \cancel{x+y} \\
 \enclose{verticalstrike}{\frac{xy}{y}} & \frac{x}{\cancel{y}} \\
 \enclose{updiagonalstrike}{x+y} & \canceluparrow{x+y} \\
 \enclose{downdiagonalstrike}{x+y} & \cancelrightarrow{x+y} \\
 \enclose{horizontalstrike,updiagonalstrike}{x+y} & \canceluparrow{x+y}
 \end{array}$$

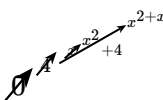
`\enclose` can also produce enclosing boxes, circles, and other notations; see [MathML `enclose` documentation](#)^[1] for a complete list.

It is worth noting that MathJax should *not* be used for formatting non-mathematical text. The preferred way for striking out text is to use the HTML strikethrough tag, `<s>[text to be stricken]</s>`, which renders as `{text to be stricken}`.

[1] <https://developer.mozilla.org/en-US/docs/Web/MathML/Element/menclose>

(25) Can I use `\enclose{counterstrike}`? :P - **Akiva Weinberger**

(64) That sneaky $19/95 = 1/5$. Nice one! - **Darth Geek**

(29) I see you can further resolve existing resolutions,  - **alan2here**

(2) Is `enclose` a *L^AT_EX* package, or only a MathML option? - **Tim Thayer**

(3) Here is a related post on meta: [Striking out equations](#). - **Martin Sleziak**

Nice command to make an arrow: `\cancelto{}{}` - **user803596**

In fact, strikethrough markup should be avoided even in text as far as possible, in the interest of accessibility: they are not picked up by screen readers (see veroniiiica.com/2020/05/29/...). CC @XanderHenderson - **The Amplitwist**

7

[+134] [2014-03-14 19:41:50] Américo Tavares

Additional decorations

`\overline`: $\overline{A}$ $\overline{AA}$ $\overline{AAA}$

`\underline`: $\underline{B}$ $\underline{BB}$ $\underline{BBB}$

`\widetilde`: $\widetilde{C}$ $\widetilde{CC}$ $\widetilde{CCC}$

`\widehat`: $\widehat{D}$ $\widehat{DD}$ $\widehat{DDD}$

`\fbox`: $\boxed{E}$ $\boxed{EE}$ $\boxed{EEE}$

$\backslash\underleftarrow{\text{F FF FFF}}$ variant: $\backslash\overleftarrow{\text{F FF FFF}}$ $\backslash\overleftarrow{\text{F FF FFF}}$

$\backslash\underrightarrow{\text{G GG GGG}}$ variant: $\backslash\overrightarrow{\text{G GG GGG}}$ $\backslash\overrightarrow{\text{G GG GGG}}$

$\backslash\underleftrightharpoonup{\text{H HH HHH}}$ $\backslash\overleftrightharpoonup{\text{H HH HHH}}$

$\backslashoverrightarrow{\text{AB ABAB ABABAB}}$ $\backslashoverleftarrow{\text{AB ABAB ABABAB}}$

$\backslashoverbrace{(n-2) + (n-1) + (n+0) + (n+1) + (n+2)}$

$\backslashunderbrace{(n-2) + (n-1) + (n+0) + (n+1) + (n+2)}$

$\backslashunderbrace{\text{underbraces can be nested, like this: } (n-2) + (n-1) + \underbrace{(n+0) + (n+1) + (n+2)}}_{\text{underbrace}}$

$\backslashoverbrace$ and $\backslashunderbrace$ accept a superscript or a subscript, respectively, to annotate the brace. For example, $\backslashunderbrace{a \cdot a \cdots a}_{b \text{ times}}$ is

$$\underbrace{a \cdot a \cdots a}_{b \text{ times}}$$

Note: $\backslashvarliminf$: $\varliminf$ and $\backslashvarlimsup$: $\varlimsup$ have special symbol of their own.

Single character accents

$\backslashcheck$: $\check{I}$

$\backslashacute$: $\acute{J}$

$\backslashgrave$: $\grave{K}$

$\backslashvec$: $\vec{u}$ $\overrightarrow{AB}$ (c.f. $\backslashoverrightarrow$ above)

$\backslashbar$: $\bar{z}$

$\backslashhat$: $\hat{x}$

$\backslashtilde$: $\tilde{x}$

$\backslashdot$ $\backslashddot$ $\backslashdddot$: $\dot{x}$, $\ddot{x}$, $\dddot{x}$

$\backslashmathring$: $\mathring{A}$

General stacking

If you cannot find your symbol remember that you can stack various symbols using

$\backslashoverset{\text{above}}{\text{level}}\text{: } ABC \overset{x^2}{\underset{\text{level}}{\longrightarrow}} T$

$\backslashunderset{\text{below}}{\text{level}}\text{: } ABC \underset{x^2}{\overset{\text{level}}{\longrightarrow}} T$

You can use these together too. You can type $X \xrightarrow[a]{a} Y$ with `X\overset{a}{\underset{b}{\to}}Y`.

Arc over points

`\overset{\huge\frown}{PQ}`: $\widehat{PQ}$ denotes the arc over points P and Q (As per comment of @Calvin Khor to @Paul Sinclair's question)

Instead of using `\fbox`, you could also use `\boxed{...}` - **Mr Pie**

(2) added arrows with text variants, some new single char accents and general stacking section. - **zwim**

(5) `stackrel` also seems to work well, as in `\stackrel{\text{def}}{=} \longrightarrow` - **Reb.Cabin**

Is there a way to do arcs over points, such as to indicate the arc of a curve between two points P and Q ? `\widearc{PQ}` doesn't seem to work. - **Paul Sinclair**

$\frown PQ$ kind of. - **user645636**

(1) The best I've been able to come up with is `\overset{\frown}{PQ}`: $\widehat{PQ}$. But since `\frown` doesn't adjust in size, it doesn't look right. Does anyone know how get a properly sized arc? - **Paul Sinclair**

(2) @PaulSinclair I offer the following `\overset{\frown}{AB}\overset{\large\frown}{CD}\overset{\Large\frown}{EF}\overset{\huge\frown}{GH}\overset{\Huge\frown}{ABC}`

$\widehat{ABC}\widehat{DEF}\widehat{G}\widehat{H}\widehat{ABC}$ - **Calvin Khor**

8

[+134] [2014-09-23 20:07:20] Lehs

Commutative diagrams

(For more examples, see [this meta question](#) ^[1].)

AMScd diagrams must start with a "require":

```
\require{AMScd}$
\begin{CD}
A @>a>> B\\
@V b V V = @VVV c V\\
C @>>d> D
\end{CD}
```

to get this diagram:

$$\begin{array}{ccc} A & \xrightarrow{a} & B \\ b \downarrow & = & \downarrow c \\ C & \xrightarrow{d} & D \end{array}$$

@>>> is used for arrow right

@<<< is used for arrow left

@VVV is used for arrow down

@AAA is used for arrow up

@= is used for horizontal double line

@| is used for vertical double line

@. is used for no arrow

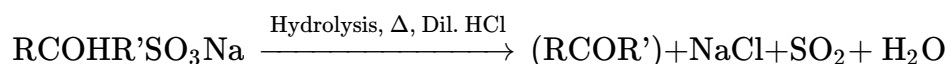
Another example:

```
\begin{CD}
A @>>> B @>{\text{very long label}}>> C \\
@. @AAA @| \\
D @= E @<<< F
\end{CD}
```

$$\begin{array}{ccccc} A & \longrightarrow & B & \xrightarrow{\text{very long label}} & C \\ & & \uparrow & & \parallel \\ D & \longleftarrow & E & \longleftarrow & F \end{array}$$

Long labels increase the length of the arrow and in this version also automatically increase corresponding arrows.

```
\require{AMScd}$
\begin{CD}
RCOHR'SO_3Na @>{\text{Hydrolysis, $\Delta$, Dil.HCl}}>> (RCOR')+NaCl+SO_2+ H_2O
\end{CD}
```



[1] <https://math.meta.stackexchange.com/q/2324/>

$$\text{RCOHR'SO}_3\text{Na} \xrightarrow{\text{Hydrolysis, } \Delta, \text{ Dil.HCl}} (\text{RCOR}') + \text{NaCl} + \text{SO}_2 + \text{H}_2\text{O}$$

Why does this code not give the correct output? - **Quark**

@Quark: The main error was a missing bracket after HCl. - **Lehs**

@Lehs Thanks. That was a silly mistake :| What if I wanted to write something below the arrow? Also, could you suggest some online website to learn MathJax? - **Quark**

@Quark: then you move a > sign: $\text{RCOHR'SO}_3\text{Na} \xrightarrow{\text{Hydrolysis, } \Delta, \text{ Dil.HCl}} (\text{RCOR}') + \text{NaCl} + \text{SO}_2 + \text{H}_2\text{O}$ I learn MathJax from the examples i.e. in this tread. - **Lehs**

@Lehs Why did you rollback my edit...? You removed the formatting from the list, broke again (in Chrome) the example diagram, and reinserted your duplicate example. Why? - **Najib Idrissi**

@NajibIdrissi: because your edit appeared as a mess in IE. The diagram wasn't even written out. Maybe there is something wrong in your or in my web-program. Now it looks good in IE. - **Lehs**

@NajibIdrissi: Now it also looks good in Chrome for Windows and for Android, plus Safari for Androids. I don't know what the problem is with the current version. - **Lehs**

(3) I realize this thread is quite old, but what about diagonal arrows? - **A. Thomas Yerger**

(2) @AlfredYerger: there are no such possibilities in AMScd. - **Lehs**

(3) @AlfredYerger Maybe presheaf can help there? See also answer and suggestions about this here: [How to draw a commutative diagram?](#) - **Martin Sleziak**

how to draw a double arrow pointing both directions to the left as well as to the right between two points in commutative diagram? - **Uncool**

How does one draw a curved arrow like in quiver? - **Tyma Gaidash**

9

[+127] [2012-10-03 11:47:44] Américo Tavares

System of equations

- Use $\begin{array}{l} \dots \end{array}$ and $\left\{ \dots \right\}$. For example, you get this:

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

by writing this:

```


$$\left\{ \begin{array}{c} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{array} \right.$$


```

- Alternatively we can use `\begin{cases}...\end{cases}`. The same system

$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

is produced by the following code

```


$$\begin{cases} a_1x + b_1y + c_1z = d_1 \\ a_2x + b_2y + c_2z = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$


```

- To align the = signs use `\begin{aligned}...\end{aligned}` and `\left\{...\right.` (see asmeurer's comment)

$$\begin{cases} a_1x + b_1y + c_1z = d_1 + e_1 \\ \quad \quad \quad a_2x + b_2y = d_2 \\ a_3x + b_3y + c_3z = d_3 \end{cases}$$

whose code is

```


$$\left\{ \begin{aligned} a_1x + b_1y + c_1z &= d_1 + e_1 \\ a_2x + b_2y &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{aligned} \right.$$


```

- To align the = signs and the terms as in

$$\begin{cases} a_1x + b_1y + c_1z &= d_1 + e_1 \\ a_2x + b_2y &= d_2 \\ a_3x + b_3y + c_3z &= d_3 \end{cases}$$

use array with l (for "align **left**"; there are also c and r) parameters

```


$$\left\{ \begin{array}{ll} a_1x + b_1y + c_1z &= d_1 + e_1 \end{array} \right.$$


```

```

a_2x+b_2y &=d_2 \\
a_3x+b_3y+c_3z &=d_3
\end{array}
\right.
$$

```

- Vertical space between equations. As explained in [Definition by cases](#) ^[1] to get a larger vertical space between equations we can use `\\[2ex]` instead of `\\`. The system

$$\begin{cases} a_1x + b_1y + c_1z = \frac{p_1}{q_1} \\ a_2x + b_2y + c_2z = \frac{p_2}{q_2} \\ a_3x + b_3y + c_3z = \frac{p_3}{q_3} \end{cases}$$

is generated by the following code

```

$$\begin{cases} a_1x+b_1y+c_1z=d_1 & \\ a_3x+b_3y+c_3z=d_3 \end{cases} \quad \begin{cases} a_2x+b_2y+c_2z=d_2 \end{cases}
\end{cases}
$$

```

in comparison with

$$\begin{cases} a_1x + b_1y + c_1z = \frac{p_1}{q_1} \\ a_2x + b_2y + c_2z = \frac{p_2}{q_2} \\ a_3x + b_3y + c_3z = \frac{p_3}{q_3} \end{cases}$$

whose code is

```

$$\begin{cases} a_1x+b_1y+c_1z=\frac{p_1}{q_1} \\ a_3x+b_3y+c_3z=\frac{p_3}{q_3} \end{cases} \quad \begin{cases} a_2x+b_2y+c_2z=\frac{p_2}{q_2} \end{cases}
\end{cases}
$$

```

- In response to [elect's comment](#) ^[2]. The following code

```

$$ \left\{ \begin{array}{l} 0 = c_x - a_{x0} - d_{x0} \frac{(c_x - a_{x0}) \cdot d_{x0}}{\|d_{x0}\|^2} + c_x - a_{x1} - d_{x1} \frac{(c_x - a_{x1}) \cdot d_{x1}}{\|d_{x1}\|^2} \\ 0 = c_y - a_{y0} - d_{y0} \frac{(c_y - a_{y0}) \cdot d_{y0}}{\|d_{y0}\|^2} + c_y - a_{y1} - d_{y1} \frac{(c_y - a_{y1}) \cdot d_{y1}}{\|d_{y1}\|^2} \end{array} \right.
\end{array}
\right.
$$

```

produces

$$\begin{cases} 0 = c_x - a_{x0} - d_{x0} \frac{(c_x - a_{x0}) \cdot d_{x0}}{\|d_{x0}\|^2} + c_x - a_{x1} - d_{x1} \frac{(c_x - a_{x1}) \cdot d_{x1}}{\|d_{x1}\|^2} \\ 0 = c_y - a_{y0} - d_{y0} \frac{(c_y - a_{y0}) \cdot d_{y0}}{\|d_{y0}\|^2} + c_y - a_{y1} - d_{y1} \frac{(c_y - a_{y1}) \cdot d_{y1}}{\|d_{y1}\|^2} \end{cases}$$

[1] <https://math.meta.stackexchange.com/a/5025/752>

[2] https://math.meta.stackexchange.com/questions/5020/mathjax-basic-tutorial-and-quick-reference/6267#comment79746_6267

(6) Is it possible to rotate text? To have a vertical word written in front of the large curly bracket that spans over all the equations? - **Steeven**

(4) @Steeven Go here [→ math.meta.stackexchange.com/questions/27798/...](https://math.meta.stackexchange.com/questions/27798/...) - **Mr Pie**

(1) Thank you, @user477343. This would be a useful feature on this list. - **Steeven**

[+125] [2013-06-29 13:58:59] Vlad K.

Colors

Named colors are browser-dependent; if a browser doesn't know a particular color name, it may render the text as black. The following colors are standard in HTML4 and CSS2 and should be interpreted the same by most browsers:

<code>\color{black}{text}</code>	<i>text</i>
<code>\color{gray}{text}</code>	<i>text</i>
<code>\color{silver}{text}</code>	<i>text</i>
<code>\color{white}{text}</code>	<i>text</i>
<code>\color{maroon}{text}</code>	<i>text</i>
<code>\color{red}{text}</code>	<i>text</i>
<code>\color{yellow}{text}</code>	<i>text</i>
<code>\color{lime}{text}</code>	<i>text</i>
<code>\color{olive}{text}</code>	<i>text</i>
<code>\color{green}{text}</code>	<i>text</i>
<code>\color{teal}{text}</code>	<i>text</i>
<code>\color{aqua}{text}</code>	<i>text</i>
<code>\color{blue}{text}</code>	<i>text</i>
<code>\color{navy}{text}</code>	<i>text</i>
<code>\color{purple}{text}</code>	<i>text</i>
<code>\color{fuchsia}{text}</code>	<i>text</i>

HTML5 and [CSS 3](#) ^[1] define [an additional 124 color names that will be supported on many browsers](#) ^[2].

Math Stack Exchange's default style uses a light-colored page background, so avoid using light colors for text. Stick to darker colors like maroon, green, blue, and purple, and remember also that 7–10% of men are color-blind and have difficulty distinguishing red and green. (Some people have difficulty distinguishing other colors too, so don't rely on colors saying "the blue part" over and over again.)

The color may also have the form `#rgb` where r, g, b are in the range 0–9, a–f and represent the intensity of red, green, and blue on a scale of 0–15, with a=10, b=11, ... f=15. For example:

<code>#000</code>	<i>text</i>	<code>#00F</code>	<i>text</i>
<code>#0F0</code>	<i>text</i>	<code>#0FF</code>	<i>text</i>
<code>#F00</code>	<i>text</i>	<code>#F0F</code>	<i>text</i>
<code>#FF0</code>	<i>text</i>	<code>#FFF</code>	<i>text</i>

#000	<i>text</i>	#005	<i>text</i>	#00A	<i>text</i>	#00F	<i>text</i>
#500	<i>text</i>	#505	<i>text</i>	#50A	<i>text</i>	#50F	<i>text</i>
#A00	<i>text</i>	#A05	<i>text</i>	#A0A	<i>text</i>	#A0F	<i>text</i>
#F00	<i>text</i>	#F05	<i>text</i>	#F0A	<i>text</i>	#F0F	<i>text</i>
#080	<i>text</i>	#085	<i>text</i>	#08A	<i>text</i>	#08F	<i>text</i>
#580	<i>text</i>	#585	<i>text</i>	#58A	<i>text</i>	#58F	<i>text</i>
#A80	<i>text</i>	#A85	<i>text</i>	#A8A	<i>text</i>	#A8F	<i>text</i>
#F80	<i>text</i>	#F85	<i>text</i>	#F8A	<i>text</i>	#F8F	<i>text</i>
#0F0	<i>text</i>	#0F5	<i>text</i>	#0FA	<i>text</i>	#0FF	<i>text</i>
#5F0	<i>text</i>	#5F5	<i>text</i>	#5FA	<i>text</i>	#5FF	<i>text</i>
#AF0	<i>text</i>	#AF5	<i>text</i>	#AFA	<i>text</i>	#AFF	<i>text</i>
#FF0	<i>text</i>	#FF5	<i>text</i>	#FFA	<i>text</i>	#FFF	<i>text</i>

You can have a look [here](#) for quick reference on colors in HTML ^[3].

[1] http://en.wikipedia.org/wiki/Cascading_Style_Sheets#CSS_3

[2] <http://www.w3.org/TR/css3-color/#svg-color>

[3] http://www.w3schools.com/html/html_colors.asp

(12) We should add that colors can be used on items other than text, such as variables and operators. The '\color' command applies to the next item: surround anything longer with braces. - **Rory Daulton**

(1) One can in fact use any CSS-compatible colour specification here, including rgb, rgba, hsl, and hsla colours. (I'd edit the answer, but have no time now. Maybe later, if nobody else beats me to it.) - **Harald Hanche-Olsen**

Can colors be used on striking out text and equations. If so, how? - **ThankYouForFlyingRyanair**

11

[+107] [2013-10-31 10:21:25] Tobias Kienzler

Tags & References

For longer calculations (or referring to other post's results) it is convenient to use the tagging/labelling/referencing system. To tag an equation use `\tag{yourtag}`, and if you want to refer to that tag later on, add `\label{somelabel}` right after the `\tag`. It is not necessary that `yourtag` and `somelabel` are the same, but it usually is more convenient to do so:

`$$ a := x^2 - y^3 \tag{*}\label{*} $$`

$$a := x^2 - y^3 \tag{*}$$

In order to refer to an equation, just use `\eqref{somelabel}`

`$$ a + y^3 \stackrel{\eqref{*}}{=} x^2 $$`

$$a + y^3 \stackrel{(*)}{=} x^2$$

or `\ref{somelabel}`

Equations are usually referred to as `\eqref{*}`, but you can also use `\ref{*}`.

Equations are usually referred to as `(*)`, but you can also use `*`.

As you can see, references are even turned into hyperlinks, which you can use externally as well, e.g. [like this](#) ^[1]. Note that you can also reference labels in other posts as long as they appear on the same site, which is especially useful when referring to a question with multiple equations, or when commenting on a post.

~~Due to a bug blocks containing a `\label` will break in preview^[2], as a workaround you can put `\def\label#1{}\$` in your post while editing and remove that on submission^[3]—unfortunately this means you won't spot misspelled references before submitting... **Just don't forget to remove that `\def` again**~~

[1] <https://math.stackexchange.com/q/537284/#mjax-eqn-e>

[2] <http://meta.math.stackexchange.com/q/11392/163>

[3] http://meta.math.stackexchange.com/questions/11392/mathjax-preview-broken-when-equations-contain-labels#comment43134_11392

(12) Also works in comments: `\eqref{*}` yields a clickable (*) - **Tobias Kienzler**

To enable automatically tagging your equations with incremental numbers, add `<script type="text/x-mathjax-config">MathJax.Hub.Config({TeX: { equationNumbers: {autoNumber: "all"} }}); </script>` to your header. - **Gerald Senarclens de Grancy**

@GeraldSenarclensdeGrancy That would however yield a global numbering on all answers to one question, not per answer. And it would break the current expectation of by default not having tags despite using unstarred `\begin{align}` etc.... Though personally I'd agree with this - **Tobias Kienzler**

(7) I'm just curious, is there a way to have the tags on the *left side* of the equation? Something like

$$(1) \qquad \sum_j k$$

But the (1) tag is all the way to the left. - **Crescendo**

How do we write a tag without brackets (because I want to tag a little square as a box of accomplishment)? - **Mr Pie**

@user477343 No idea, you could ask at tex.stackexchange.com or see if the manual of amsmath has something... - **Tobias Kienzler**

@Crescendo You could ask on tex.stackexchange.com for a solution. A workaround could be `\begin{array}{lc}` or similar. - **Tobias Kienzler**

(3) Hey, I figured how to tag without brackets. You simply put what is inside the braces: `{|tag*{. . .}}` which I learnt from here → math.meta.stackexchange.com/questions/27731/... - **Mr Pie**

How do I add several tags analogously to `empheq`? e.g.

$$\begin{cases} 2x_1 + 3x_2 = 8 & (1a) \\ 7x_1 + 9x_2 = -13 & (1b) \end{cases}$$

- **Dmitrii Demenev**

I can't make `\tag{*}\label{*}` work—when I use it, the equation is never rendered into an image and in the code that is displayed in place of the image this show up as `"\tag{}\label{}`". `\tag{1}\label{1}` on the other hand does work. - **HelloGoodbye**

12

[+104] [2012-08-31 19:46:55] MJD

Continued fractions

To make a continued fraction, use `\cfrac`, which works just like `\frac` but typesets the results differently:

$$x = a_0 + \frac{1^2}{a_1 + \frac{2^2}{a_2 + \frac{3^2}{a_3 + \frac{4^4}{a_4 + \dots}}}}$$

Don't use regular `\frac` or `\over`, or it will look awful:

$$x = a_0 + \frac{1^2}{a_1 + \frac{2^2}{a_2 + \frac{3^2}{a_3 + \frac{4^2}{a_4 + \dots}}}}$$

You can of course use `\frac` for the compact notation:

$$x = a_0 + \frac{1^2}{a_1 + \frac{2^2}{a_2 + \frac{3^2}{a_3 + \frac{4^2}{a_4 + \dots}}}}$$

Continued fractions are too big to put inline. Display them with `$$...$$` or use a notation like $[a_0; a_1, a_2, a_3, \dots]$.

(4) The RHS of the following continued fraction

$$\frac{a_1}{b_1 + \frac{a_2}{b_2 + \frac{a_3}{b_3 + \ddots}}} = \frac{a_1}{b_1} + \frac{a_2}{b_2} + \frac{a_3}{b_3} + \dots$$

can be typeset with the `\genfrac` command '`\genfrac{}{}{}{a_1}{b_1} {\genfrac{}{}{0pt}{}{+}} {\genfrac{}{}{}{a_2}{b_2}} {\genfrac{}{}{0pt}{}{+}} {\genfrac{}{}{}{a_3}{b_3}} {\genfrac{}{}{0pt}{}{+}\dots}`' - **Américo Tavares**

I wonder if something like $\frac{1}{2} + \frac{3}{4}$ would be good enough? It is much simpler. `(\frac{12{\vphantom{1}}\atop+}{\frac{34}})` - **MJD**

(2) Yes, it is. I didn't mention it because in *User's Guide for the amsmath Package* it is written the following: "Note. For technical reasons, using the primitive fraction commands `\over`, `\atop`, `\above` in a LATEX document is not recommended (see, e.g., *amsmath.faq*)." - **Américo Tavares**

(5) Happily, we are not writing *L^AT_EX* documents here. - **MJD**

(13) Or write `\underset{j=1}{\overset{\infty}{\LARGE\mathrm{K}}}\frac{a_j}{b_j}=\cfrac{a_1}{b_1+\cfrac{a_2}{b_2+\cfrac{a_3}{b_3+\ddots}}}` to get

$$\overset{\infty}{\underset{j=1}{\mathrm{K}}} \frac{a_j}{b_j} = \frac{a_1}{b_1 + \frac{a_2}{b_2 + \frac{a_3}{b_3 + \ddots}}}$$

- **Américo Tavares**

(12) @AméricoTavares Or use `\mathop` instead of `\overset` and `\underset`: `\mathop{\LARGE\mathrm{K}}_{i=1}^{\infty} \frac{a_i}{b_i}`

$$\overset{\infty}{\underset{i=1}{\mathrm{K}}} \frac{a_i}{b_i}$$

- **AlexR**

@AméricoTavares, Why don't you edit the answer and put this extremely helpful command into there, I think that would be more helpful. - **user249332**

13

[+98] [2013-11-13 13:45:08] Abramo

Using `\newcommand`

I would like to remark that it is possible to define LaTeX commands as you do in your TeX files. I felt so happy when I first discovered it! It's enough to insert something like

```
$ \newcommand{\SES}[3]{ 0 \to #1 \to #2 \to #3 \to 0 } $
```

at the top of your post (remember the dollars!). Then you can just use your commands as you are used to do: in my example typing `$$ \SES{A}{B}{C} $$` will produce the following:

$$0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$$

It's also possible to use plain `\def`:

```
\def\ses#1#2#3{0 \to #1 \to #2 \to #3 \to 0}
```

and then `$$\ses{A}{B}{C}$$` will produce the same output.

(12) Be aware that this affects the entire post, possibly even the frontpage, so it should be used with great care. - **AlexR**

(2) $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 0$ - **ericw31415**

(7) @AlexR It's been fixed since. - **iBug**

This unfortunately doesn't work in stackedit.io which I sometimes use to compose and edit longer stackexchange posts.

Anyone aware of possible workarounds or alternatives? - **joseville**

Why does this not work with `\def`? - **Tyma Gaidash**

14

[+89] [2013-06-14 01:33:05] **leonbloy**

`\implies ( \implies )` is a marginally preferable^[1] alternative to `\Rightarrow (\Rightarrow)` for implication.

There's also `\iff \iff` and `\impliedby \Leftarrow`.

`\to (\rightarrow)` is preferable to `\rightarrow` or `\longrightarrow` for things like $f: A \rightarrow B$. The reverse is `\gets (\leftarrow)`.

[1] <https://tex.stackexchange.com/questions/47063/rightarrow-vs-implies-and-does-not-imply-symbol>

(6) Why is it preferable? - **MJD**

(19) `implies` looks nicer as the arrow is longer and `\to` is quicker to right (and it's also what you say in your head while typing it). at least that's what I think. - **John Salvatierrez**

(4) Remember the difference between `\to` and `\mapsto` as in $T: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x + 1$ produced by `T:\mathbb{R}\to\mathbb{R},\; x\mapsto x+1` - **yo'**

(9) I prefer using `\to` when it appears as part of a larger propositional formula, rather than at the top level, i.e. $p \wedge ((q \vee r) \rightarrow s)$, because the spacing is similar to that of other binary operators. `\implies` is better for sentence- or clause-level implications, or in displays, i.e.

$$x + 2 = 4 - x \implies x = 1.$$

- **Mario Carneiro**

I have always used `|Longleftarrow` for `|impliedby`. It generates the same thing anyway, for which the former generates `\Leftarrow` and the latter generates `\Leftarrow` with `|Leftarrow \Leftarrow` as an alternate for reverse implication. - **Mr Pie**

@yo' instead of `|mathbb` you could also use `|Bbb` as a matter of fact :) - **Mr Pie**

Is there a way to add some text above `\implies`? Such as "by (1)" to refer to another equation that is used for substitution and similar cases. - **Alexandros**

(2) @Alexandros yes `\overset{3.1415}{\underset{26535}{\implies}}` produces

$$\overset{3.1415}{\underset{26535}{\implies}}$$

- **user645636**

15

[+84] [2013-10-25 17:47:09] **E.P.**

Big braces

Use `\left` and `\right` to make braces - (round), `[square]` and `{curly}` - scale up to be the size of their arguments. Thus

```


$$f\left(\left[\frac{1+\left\{x,y\right\}}{\left(\frac{x}{y}+\frac{y}{x}\right)(u+1)}+a\right]^{3/2}\right)$$


```

renders as

$$f\left(\left[\frac{1+\{x,y\}}{\left(\frac{x}{y}+\frac{y}{x}\right)(u+1)}+a\right]^{3/2}\right).$$

Note that curly braces need to be escaped as `\{ \}`.

If you start a big brace with `\left` and then need to match that to a `\right` brace that's on a different line, use the forms `\right.` and `\left.` to make "shadow" braces. Thus,

```


$$a=\left(1+2+3+\cdots\right.\backslash\left.\infty-2+\infty-1+\infty\right)$$


```

renders as

$$a=(1+2+3+\cdots\cdots+\infty-2+\infty-1+\infty).$$

There is also a `\middle` construct which is useful when one has a mid-expression brace which must also scale up:

```


$$q\left|\frac{\frac{x}{y}}{\frac{u}{v}}\right|p$$


```

renders as

$$\left\langle q\left|\frac{\frac{x}{y}}{\frac{u}{v}}\right|p\right\rangle.$$

Note that constructs like `\left\langle`, `\left|` and `\left\|` are also possible.

Alternatively there also exists the `\big` hierarchy whose pairing is not mandatory, you can type

$$\big(\frac{1}{x}\big)$$

The advantage of left/right is that it dimensions automatically, but has the inconvenient of not producing consistent results depending of the vertical extension of its inner content, instead the big hierarchy has fixed size:

$$\Big(\big(\Big(\big((x)\big)\Big)\bigg)\Big)$$

(6) Note: `\Big( ... \Big)` produces $\left(\dots \right)$ but this bracket size is fixed in all situations unlike `\left( ... \right)` which varies in size with its contents. `\Big` can be useful in various situations. - **Nick**
Added a paragraph about the big hierarchy. - **zwim**

16

[+80] [2014-02-26 08:30:31] Anonymous Computer

Limits

To make a limit (like $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$), use this syntax:

First, start off with `\lim`. This renders as $\lim$. The backslash is there to prevent things like *lim*, where the letters are slanted.

Second, add `\limits_{x \rightarrow 1}` inside. The code now looks like `\lim \limits_{x \rightarrow 1}`, and renders as $\lim_{x \rightarrow 1}$. The `\to` inside makes the right arrow, rendered as $\rightarrow$. The `_` makes the $x \rightarrow 1$ go underneath the $\lim$. Finally, the pair of curly braces `{ }` makes sure that $x \rightarrow 1$ is treated as a whole object, and not two separate things.

Lastly, add the function you want to apply the limit to. To make the limit mentioned above, $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$, simply use `\lim \limits_{x \rightarrow 1} \frac{x^2-1}{x-1}`.

And that is how you make a limit using MathJax.

(29) Why not just `\lim_{x \rightarrow 1}`

$$\lim_{x \rightarrow 1}?$$

As I understand it `\limits` is only needed for operations that don't already understand limits, for example if you want to use `+` and get

$$\sum_{i=1}^k \text{ instead of } +_{i=1}^k$$

When used inline, your suggestion will produce $\lim_{x \rightarrow 1}$ instead of the more compact form $\lim_{x \rightarrow 1}$ that mathjax normally chooses. Are you sure this is good advice? - **MJD**

(5) @MJD `\lim_{x \rightarrow 1}` renders to $\lim_{x \rightarrow 1}$, and `\lim \limits_{x \rightarrow 1}` renders as $\lim \limits_{x \rightarrow 1}$. Note how the $x \rightarrow 1$ is separated from the first limit, and not directly underneath. We do not write limits like that in real life, so we use `\limits`. - **Anonymous Computer**

(2) I meant that the second limit renders to $\lim_{x \rightarrow 1}$ - **Anonymous Computer**

(10) Limits are usually written that way in typeset materials like papers and books when the limit is inline, rather than a displayed formula, and that's why MathJax typesets it that way. - **MJD**

(14) The issue with this answer is that it is trying to "force" display mode on inline code. Doing so makes the text look

less pretty. For example, see how the spacing between the lines change when I force display mode using `\lim\limits_{x\mapsto 1}\dfrac{1}{x}`. On the other hand, when I let TeX do what it wants to do, using `\lim_{x\mapsto 1}\frac{1}{x}`, the spacing between the lines stays the same, which is much neater: $\lim_{x\mapsto 1} \frac{1}{x}$. This is much easier on the eyes. If you want to make your math mode more prominent then take a new line using `$$-$$-user1729`

(9) The moral is: TeX was written by a jolly clever chap. Let it do what it wants, because it does it for a reason! - **user1729**

(2) Part 11 of the "question" shows how to write limits in the way they were meant to be written in LaTeX and MathJax. - **David K**

17

[+69] [2014-06-24 08:34:28] MvG

Arbitrary operators

If an operator is not available as a built-in command, use `\operatorname{...}`. So for things like

$$\operatorname{arsinh}(x)$$

write `\operatorname{arsinh}(x)` since `\arsinh(x)` will give an error and `arsinh(x)` has wrong font and spacing: *arsinh*(*x*).

This was already mentioned in a [comment](#) ^[1] by [Charles Staats](#) ^[2]. You might consider this an addition to the FAQ section on `\lim`, `\sin` and so on.

For operators which need limits above and below the operator, use `\operatorname*{...}`, as in

$$\operatorname*{Res}_{z=1}\left(\frac{1}{z^2-z}\right)=1$$

New operators may also be defined using the `\DeclareMathOperator` syntax: `\DeclareMathOperator{newOperatorCommand}{newOperator}` defines a new operator. On the page where this code occurs, `newOperatorCommand` will be rendered as `newOperator`.

[1] http://meta.math.stackexchange.com/questions/5020/mathjax-basic-tutorial-and-quick-reference/11638#comment19884_5020

[2] <http://meta.math.stackexchange.com/users/455/charles-staats>

(1) We can also use `\rm ...`. For example, `\rm arsinh` yields *arsinh*. - **Felix Marin**

(19) @Felix: `\rm` will change the font but not the spacing. `\operatorname{arsinh}x` renders as “*arsinh x*” while `\rm arsinh`*x* renders as “*arsinhx*”. Notice the added space between operator and operand in the first example, which is missing in the second. On the whole, I'd say that `operatorname` is a lot more in the spirit of semantic markup, declaring *what* you want to write instead of *how* you want to write it, so I'd strongly suggest using this. - **MvG**

(5) Thanks. I didn't know there was a difference between them. I always avoided `operatorname` because it was too long. - **Felix Marin**

(3) Thanks for this. I thought carefully about whether to put `\operatorname` in the main post, and decided to leave it out. The reason is simple: If a beginner omits `\operatorname`, the resulting formula will still be perfectly clear, and a more experienced user will have no trouble inserting the `\operatorname` where it is needed. So including it in the main post would not be a good use of space. - **MJD**

(3) ... I always use `"\text{operator}"`. Hmmm, *arsinh x* vs *arsinh x*. - **JP McCarthy**

(4) If you use the same operator many times, I think you can do `\DeclareMathOperator{\arsinh}{arsinh}` at the post's top. Never tried it though... - **MickG**

What is the code for the last one? - **Laxmi Narayan Bhandari**

@Laxmi you can right-click on MathJax formulas and select "Show Math As / TeX Commands" to see the code for any formula. You can also click on the date of the edits to see edit history, and in that history use "Side-by-side Markdown" rendering to see the source of the whole post. - **MvG**

18

[+68] [2016-01-10 19:48:49] webbertiger

Highlighting equation

To highlight an equation, `\bbox` can be used. E.g.,

```


$$e^x = \lim_{n \rightarrow \infty} \left( 1 + \frac{x}{n} \right)^n$$

\quad (1)

```

produces

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \quad (1)$$

By default, the bounding box is "tight", so it doesn't extend beyond the characters used in the formula. You can add a little space around the equation by adding a measurement after the color. E.g.,

```


$$e^x = \lim_{n \rightarrow \infty} \left( 1 + \frac{x}{n} \right)^n$$

\quad (1)

```

produces

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \quad (1)$$

To add a border, use

```


$$e^x = \lim_{n \rightarrow \infty} \left( 1 + \frac{x}{n} \right)^n$$

\quad (2)

```

produces

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \quad (2)$$

You can do both border and background, as well:

```


$$e^x = \lim_{n \rightarrow \infty} \left( 1 + \frac{x}{n} \right)^n$$

\quad (1)

```

produces

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n \quad (1)$$

(3) When using constructs like this, please heed the points raised in [this discussion](#) on usage of colour. - **Lord_Farin**

(2) This would be a very helpful feature. - **user379641**

19

[+63] [2014-06-24 08:39:43] MvG

Absolute values and norms

The absolute value of some expression can be denoted as `\lvert x\rvert` or, more generally, as `\left\lvert ... \right\rvert`. It renders as $|x|$.

The norm of a vector (or similar) can be denoted as `\lVert v\rVert` or, more generally, as `\left\lVert ... \right\rVert`. It renders as $\|v\|$. (You may also write `\left| ... \right|` instead.)

In both cases, the rendering is better than what you'd get from `|x|` or `||v||`, which render with bars that don't descend low enough and sub-optimal spacing. At least on some browsers, so here is a screenshot how it looks for me, using Firefox 31 on OS X:

$$|x|, \|v\| \longrightarrow \lvert x \rvert, \lVert v \rVert$$

And here is the same formula rendered by your browser:

$$|x|, ||v|| \longrightarrow |x|, \|v\|$$

It was typeset as

`$$|x|, ||v|| \quad\quad\quad\longrightarrow\quad\quad \lvert x \rvert, \lVert v \rVert$$`

(9) You can use `\|x\|` instead of `\lVert x \rVert`; $\|x\|$ and $\lVert x \rVert$. (I don't think that there is a difference between them. I've tried [asking on SE](tex.stackexchange.com/questions/77767/whats-the-correct-way-to-write-norm)). - **Martin Sleziak**

On my browser $|x|$ and $\lvert x \rvert$ ($|x|$ and $\lvert x \rvert$) look identical, contrary to your claim. Perhaps you need to show an example more complicated than just 'x'? - **MJD**

@MJD: What's your browser? I included a screenshot to support my claim. - **MvG**

Usually various versions of Firefox on either Linux or Windows. I happen to have Windows 8 booted now, so here's a screenshot from there: a.pomf.se/jrujkq.PNG The bar height looks good on both pairs of symbols; the spacing is a little off for the `||` version. On Linux they looked the same. - **MJD**

Here's a screenshot with FF 31.0 under Linux: a.pomf.se/fhwmjo.png - **MJD**

(4) The difference in output that you are seeing has to do with whether you have the STIX fonts installed locally on your computer or not. The `|` in STIX doesn't descend below the baseline, while in the MathJax TeX fonts it does. - **Davide Cervone**

"`\lvert x\rvert` or, more generally, as `\left\lvert ... \right\rvert`" What do you mean by "or more generally"? Why would one use one vs the other? - **Don Hatch**

@DonHatch The `\left` and `\right` usually affects the following parentheses, brackets and such and they make them scale with the size of the enclosed expression. So I assume in this case the general version also scales with content. You may want to use it for fractions, integrals, matrices and other large objects. The only benefit of the less general version is its shorter code. - **Lukáš J.**

20

[+59] [2015-08-27 14:14:37] David K

Giving reasons on each line of a sequence of equations

To produce this:

$$v + w = 0 \qquad \text{Given} \qquad (1)$$

$$-w = -w + 0 \qquad \text{additive identity} \qquad (2)$$

$$-w + 0 = -w + (v + w) \qquad \text{equations (1) and (2)}$$

write this:

```
\begin{align}
v + w &= 0 && \text{\text{Given}} \tag{1} \\
-w &= -w + 0 && \text{\text{additive identity}} \tag{2} \\
-w + 0 &= -w + (v + w) && \text{\text{equations (1) and (2)}}
\end{align}
```

(1) Using multiple `\tag` commands in my equations causes them to break. It only takes one tag per equation and it labels the entire thing instead of allowing tagging on a *per-line* basis. Any ideas? - **code_dredd**

@code_dredd The particular formatting in this answer still seems to work. Perhaps you could post your formulas in a new meta question to get help with them. - **David K**

Why would you use `\tag`, instead of just using `()`? - **Some Guy**

(1) @SomeGuy First, this is what `\tag` is meant for. It puts the (1) exactly where it should be, at the right margin. Second, if you have an equation like $0 = ax^2 + bx + c$ and you **just** use (1) on it instead of `\tag{1}`, you end up with $0 = ax^2 + bx + c(1)$. To fix this, instead of `\tag` you have to insert other commands to make enough blank space. Not a net gain, in my opinion. - **David K**

@DavidK but I couldn't create a space between them - **Andre Lin**

@AndreLin Space between what and what? - **David K**

@DavidK Between the equation and (1) - **Andre Lin**

@AndreLin That's strange. I sometimes don't see the (1) at all if the page is loading too slowly, but when I do see it it's as far to the right as any of the math or text gets, which is as much space from the rest of the equation as you can get. - **David K**

@DavidK I don't think your situation is common - **Andre Lin**

@DavidK Isn't it a WiFi issue - **Andre Lin**

@AndreLin In any case `\tag` is a commonly-used LaTeX/MathJax command, for example [here](#). - **David K**

21

[+48] [2016-01-28 05:03:21] David

Pack of cards

If you are asking (or answering) a combinatorics question involving packs of cards you can make it look more elegant by using `\spadesuit`, `\heartsuit`, `\diamondsuit`, `\clubsuit` in math mode:



Or if you're really fussy:

`\color{red}{\heartsuit}` and `\color{red}{\diamondsuit}`



You can also enter the standard Unicode characters (U+2660 BLACK SPADE SUIT etc.) literally, or copy them from [here](#):

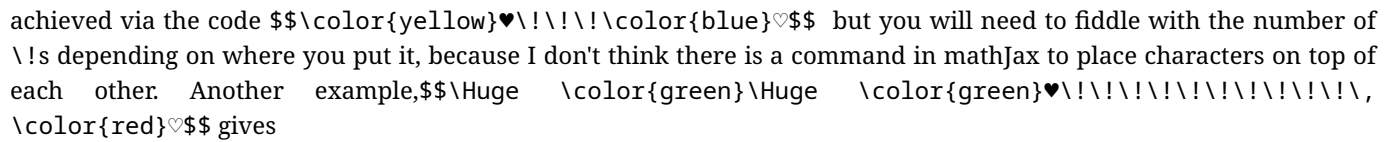


(1) This is very nice! Is there other auto-shapes or stickers? - **user379641**

(1) Is it also possible to draw the spade and club in outlines and fill the heart and diamond with a colour? - **user379641**

(2) @AlwaysConfused None that come to mind. Google search turned up [this](#) which might help. Otherwise search for a TeX/LaTeX/MathJax symbol table. - **David**

To the above commenters - it is possible, for instance



22

23

10/03/26, 16:28

Note that comments below indicate that on some configurations at least, $^\circ$ renders inferior to $^\circ$. And I recently had [a post of mine edited](#) ^[3] just for the sake of turning $^\circ$ into $^\circ$, indicating that someone felt rather strongly about this. So the suggestion above does seem somewhat controversial at the moment. I maintain that from a semantic point of view, $^\circ$ is superior to $^\circ$, and if the rendering suffers from this, then it's a bug in MathJax. After all, LaTeX offers a proper degree symbol in the tex companion fonts, indicating that someone there, too, decided that $^\circ$ is not perfect. But if things are broken now, I can't fault people from pragmatically sticking with the rendering they prefer. Personally I prefer semantics, also for the sake of screen readers.

Accessibility

Aside from appearance, one consideration in choosing which notation to use is how it will get parsed by screen readers. For example, [ChromeVox](#) ^[4] reads both 45° and 45° as "forty-five degrees", while the other two are pronounced as "forty-five oh", which may be a reason to avoid them.

Usepackage

Commonly in Latex you can `\usepackage{gensymb}` to get the `\degree` symbol, however on Stack Exchange this is not an option. Note that even if you can do this it will typically affect the entire page, which may have side effects for other users. So don't rely on this approach.

[1] http://en.wikipedia.org/wiki/Degree_symbol

[2] http://en.wikipedia.org/wiki/Degree_symbol#Keyboard_entry

[3] <https://math.stackexchange.com/posts/612972/revisions#rev-arrow-2c001335-10cd-45c9-a20f-6458a8020593>

[4] <http://www.chromevox.com/>

(2) If mathjax loads siunitx or gensymb, there is then `\degree` in latex which is the degree symbol. - **dustin**

(1) @dustin: I couldn't find siunitx or gensymb mentioned anywhere in the MatJax source repository. Are they available as some kind of third-party extension? If so, where? Since MathJax is *not* LaTeX, packages can't be loaded unless they have been migrated. By the way, all occurrences of "degree" in the MathJax sources refer to something else, as far as I can tell, so there really doesn't seem to be a `\degree` macro. There should be one, imho. - **MvG**

(2) I am not a mathjax expert. I just know latex. I just gave that suggestion in case they were available. [Siunitx](#) would be a great package to have. If you aren't familiar, you will see the advantage by scanning the documentation on ctan. - **dustin**

(16) On my display, $^\circ$ looks bad and $^\circ$ looks good: a.pomf.se/xnlfyg.png - **MJD**

(3) Degree sign can generally be typed by holding down Alt and typing 0176 on the numeric keypad. $^\circ$ (I don't know how international the actual number is). The leading zero is required. - **Joffan**

(1) @Joffan: 167 is the decimal representation of the Codepoint for $^\circ$ in Latin 1, Unicode and CP-1252. Without the leading zero, CP-437 gets applied instead, at least in typical English-speaking countries, so you'd use Alt+248 there. The Wikipedia article I linked to already describes those two ways of entering the symbol, and en.wikipedia.org/wiki/Alt_code has some more details. - **MvG**

How to use [Radian \(c\)](#), [gradian \(g\)](#) and [Steradian \(sr\)](#)? And also, Angstrom (though a lenght unit)? - **user379641**

Actually we can write degrees by 90° (O for Orange, using lowercase o, like 'o'), and it'd render it close to degrees symbol

$$90^\circ + 30^\circ + 45^\circ$$

- **user427802**

(1) @AbhasKumarSinha It looks quite slanty to me. - **Tom Hale**

(3) @StephenG: I'm not happy with [your latest edit](#). I feel that it is not helpful to users if we suggest even more ways to poorly format that symbol (like $^\circ$ imho), or to mention a LaTeX approach just to say it won't work. You deleted the example for 45° , but kept the sentence talking about it, including the colon. I'm reluctant to revert your edit on a CW page without a conversation, but as it stands I see the edit as a change for the worse. Can we find a combined solution? - **MvG**

(1) I just wrote a [feature request](#) for a `\degree` symbol, since I believe it would be technically easy and conceptually beneficial to have such a symbol defined for the whole site. - **MvG**

@MvG I have added an entry to the "renders as" table for keyboard entry (which frankly looks awful IMO) but

regarding your "unhappiness" note only one line was deleted from the version preceding my first edit and I regard your belief that this justifies your claim my edit was "unhelpful" is nonsense. I fail to see how undoing my edit helps anyone but you. - **StephenG - Help Ukraine**

(1) While we're at it, I included my comment on accessibility from the feature request post, since it may be more useful here. It would be nice if other people tested other screen readers to get a sample size of higher than one. - **Misha Lavrov**

I recently discovered `\mathring` and hence there is a further variant `a\mathring{}` which is neither *circa*^o nor the actual unicode symbol *a*^o - **Calvin Khor**

24

[+37] [2015-07-26 15:48:16] David K

Long division

```


$$\begin{array}{r}
 13 \\
 4 \overline{)52} \\
 \underline{4} \phantom{2} \\
 12 \\
 \underline{12} \\
 0
 \end{array}$$


```

$$\begin{array}{r}
 13 \\
 4 \overline{)52} \\
 \underline{4} \\
 12 \\
 \underline{12} \\
 0
 \end{array}$$

One important trick shown here is the use of `\phantom{2}` to make a blank space that is the same size and shape as the digit 2 just above it.

This is adapted from <https://stackoverflow.com/a/22871404/3466415> (which uses slightly different but not less valid formatting).

(10) Synthetic division. Example to find that

$$x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6) + 0$$

$$\begin{array}{r|rrrr}
 & x^3 & x^2 & x^1 & x^0 \\
 & 1 & -6 & 11 & -6 \\
 \textcolor{red}{1} & \downarrow & 1 & -5 & 6 \\
 \hline
 & 1 & -5 & 6 & | \textcolor{blue}{0}
 \end{array}$$

```

\begin{array}{c|rrrr}
& x^3 & x^2 & x^1 & x^0 \\
& 1 & -6 & 11 & -6 \\
\textcolor{red}{1} & \downarrow & 1 & -5 & 6 \\
\hline
& 1 & -5 & 6 & | \textcolor{blue}{0}
\end{array}

```

Américo Tavares

(1) @Maria Mazur For the same example $\frac{x^3 - 6x^2 + 11x - 6}{x - 1} = x^2 - 5x + 6$:

x^3	$-6x^2$	$+11x$	-6	$x - 1$
$-x^3$	$+x^2$			$x^2 - 5x + 6$
	$-5x^2$	$+11x$	-6	
	$5x^2$	$-5x$		
		$+6x$	-6	
		$-6x$	$+6$	
		0	0	

I've used this code `\begin{array}{rrrr|l} x^3 & -6x^2 & +11x & -6 & x - 1 \\ -x^3 & +x^2 & & & x^2 - 5x + 6 \\ \hline -5x^2 & +11x & -6 & & \\ 5x^2 & -5x & & & \\ \hline & +6x & -6 & & \\ & -6x & +6 & & \\ \hline & 0 & 0 & & \end{array}` - **Américo Tavares**

25

[+31] [2016-09-23 06:02:15] Alexis Olson

Displaystyle and Textstyle

Many things like fractions, sums, limits, and integrals display differently when written inline versus in a displayed formula. You can switch styles back and forth with `\displaystyle` and `\textstyle` in order to achieve the desired appearance.

Here's an example switching back and forth in a displayed equation:

```


$$\sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2}$$


```

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2} \rightarrow \sum_{n=1}^{\infty} \frac{1}{n^2}$$

It is possible to switch style inline as well:

Compare $\lim_{t \rightarrow 0} \int_t^1 f(t) dt$ versus $\lim_{t \rightarrow 0} \int_t^1 f(t) dt$.

Compare $\lim_{t \rightarrow 0} \int_t^1 f(t) dt$ versus $\lim_{t \rightarrow 0} \int_t^1 f(t) dt$.

Do observe that the taller formulas gotten with `\displaystyle` distort the line spacing.

Filler text, more filler text and even more filler text, and an outrageous amount of filler text.

It would not occur to me to use $\lim_{t \rightarrow 0} \int_t^1 f(x) dx$ here. As we see, a formula typeset in `displaystyle` makes it necessary to move the lines further apart. A ridiculous amount of filler text to make a point. Not pleasing to the eye at all.

In other words, there is also a reason TeX defaults to `\textstyle` when typesetting inline formulas.

(4) Oh!! I was always confused on why some people had `\displaystyle`. - **Simply Beautiful Art**

(1) @SimplyBeautifulArt I was always wondering on why the math expressions of some people looked nicer than mine.. - **user486983**

There is also AbC `\scriptstyle{AbC}` and AbC `\scriptscriptstyle{AbC}`. - **emacs drives me nuts**

26

[+29] [2016-09-22 15:03:17] Workaholic

Vertical Spacing

Some formulas such as $\bar{a} + \bar{b} = \overline{a \cdot b}$, $\sqrt{a} - \sqrt{b}$, do not look quite right when it comes to vertical spacing. Fortunately, there is more than one way to fix this. One can for instance employ the `\mathstrut` command as follows:

```
\sqrt{\mathstrut a} - \sqrt{\mathstrut b}
```

Which yields: $\sqrt{a} - \sqrt{b}$. Or using `\vphantom` (vertical phantom) command, which measures the height of its argument and places a math strut of that height into the formula.

```
\sqrt{\vphantom{b} a} - \sqrt{b}
```

Which renders as: $\sqrt{a} - \sqrt{b}$.

Another issue is with the spacing within lines in situations like this,

Based on the previous technique, we can simplify $\frac{1}{\sqrt{a} - \sqrt{b}}$, and we thus get the result of the previous limit. [this text is added to show alignment with the above smashed object]

These two lines are too far apart, but this is unnecessary since the second line is very short. We can solve this by using the `\smash` command, to get:

Based on the previous technique, we can simplify $\frac{1}{\sqrt{a} - \sqrt{b}}$, and we thus get the result of the previous limit. [this text is added to show alignment with the above smashed object]

Beware - as above - the smashed text may overlap the next line if that line extends far enough to reach the smashed object, so this solution is not always feasible (it is esp. likely to occur in slim-width browsers, e.g. phones). Analogous overlapping may occur with any prior lines. Note that `smash` can be restricted to top or bottom with an argument: `\smash[t]...` or `\smash[b]...`

Alternatively, one can also sneak in a rule of zero width `\rule{0pt}{2ex}`, as [explained here](#). - **on4aa**

(4) On Android, at least, the results of `\smash` look awful. The formula overlaps the text. - **dfeuer**

@dfeuer Android is not the source of the problem. Rather it is the fact that the browser window width is so short that the 2nd line extends far enough to be below the radical, so smashing the radical causes it to overlap the 2nd line. The same thing occurs in any browser if you make its width small enough. - **Bill Dubuque**

@BillDubuque, okay, but it's still a problem for mobile browsers. - **dfeuer**

@dfeuer Again, it's not "mobile" browsers that are the source of the problem - rather it is the fact that there is something displayed below the smashed object. That is more likely to occur on phones since they are more likely to have shorter width screens. If you shrink the width of a desktop browser window to be small enough (or extend the length of the following line) then the above answer will show the problem there too (e.g. I edited the answer to show that). - **Bill Dubuque**

27

[+29] [2018-01-26 02:01:28] **jdhaio**

Equation numbering

Simple equation

To give an equation a number, use the `\tag{}`. To refer to it later, use `\label{}` to label this equation. When you want to refer to it, use `\eqref{}`. For example,

$$e = mc^2 \tag{1}$$

Equation (1) is one of the greatest equations in mankind's history. Equation (1) is produced using the following code,

```
$$e=mc^2 \tag{1}\label{eq1}$$
```

To refer to it, use `\eqref{eq1}`.

Multi-line equation

Multi-line equation is actually just one equation rather than several equations. So the correct environment is `aligned` instead of `align`.

$$\begin{aligned} a &= b + c \\ &= d + e + f + g \\ &= h + i \end{aligned} \tag{2}$$

Equation (2) is a multi-line equation. The code to produce equation (2) is

```
$$\begin{equation}\begin{aligned} a &= b + c \\ &= d + e + f + g \\ &= h + i \end{aligned}\end{equation}\tag{2}\label{eq2}$$
```

Multiple aligned equations

For multiple aligned equations, we use the `align` environment.

$$a = b + c \tag{3}$$

$$x = yz \tag{4}$$

$$l = m - n \tag{5}$$

Equation (3), (4) and (5) are multiple equations aligned together. The code to produce these equations is,

```
$$\begin{align} a &= b + c \tag{3}\label{eq3} \\ x &= yz \tag{4}\label{eq4} \\ l &= m - n \tag{5}\label{eq5} \end{align}$$
```

I don't believe there is any difference between *align* and *aligned*, but whatever feels comfortable I suppose. - **Mr Pie**

There is actually a difference, read [here](#) for a detailed discussions. - **jdhao**

thank you very much for clearing up that understanding :) - **Mr Pie**

You are welcome. When in doubt, always google it first :). - **jdhao**

If there's an equation with multiple lines, is there a way to add tags on a *per-line* basis, i.e. `\tag{1}` for line 1, `\tag{2}` for line 2, etc? If I use the `\tag{...}` commands, I can only use one per equation and it labels the entire equation, not each line. - **code_dredd**

I am not aware of this kind of command. What is your use case? - **jdhao**

the last equation numbering can also be used with `align*` instead of `align` - **user173262**

I get all tags on the first line: "a = b + c (3)(4)(5)". - **Jiri Kriz**

Why do we need both `$$` and `\begin{equation}` with `aligned`? I noticed that when I omitted `\begin{equation}` that I did not get an equation number, but that does not happen when I use only `$$` without any other environments inside. - **Randy Cragun**

If equations are not being numbered, is there any (practical) difference between `aligned` and `align`? (I understand they **should** be used with single equations and multiple equations respectively.) I've checked the link mentioned in an earlier comment which shows several examples of differences, but none of them seem to apply when using MathJax

here on Math Stack Exchange; they use environments like `tabular` or `list` or `enumerate`, none of which are recognized by MathJax. Also, is there any reason to use `\begin{equation}` and `\end{equation}` if equations are not being numbered? - A.J.

I found in [physics stackexchange](#), MathJax doesn't support `\label{ }` for reference. To align multiple equations, we need to escape twice like: `\begin{align} a &= b + c \tag{Eq.3} \\ x &= yz \tag{Eq.4} \\ l &= m - n \tag{Eq.5} \end{align}`. Also, you don't need `$$`, which is for a single equation and it is equivalent to `\begin{equation}` - Leon Chang

28

[+22] [2018-01-19 07:28:44] GNUSupporter 8964民主女神 地下教會

Linear programming

Formulation

A theoretical LPP can be typeset as

```
\begin{array}{ll}
\text{maximize} & c^T x \\
\text{subject to} & d^T x = \alpha \\
& 0 \leq x \leq 1.
\end{array}
```

$$\begin{aligned} & \text{maximize} && c^T x \\ & \text{subject to} && d^T x = \alpha \\ & && 0 \leq x \leq 1. \end{aligned}$$

To input a numerical LPP, use `alignat` instead of `align` to get better alignment between signs, variables and coefficients.

```
\begin{alignat}{5}
\max \quad & z = & x_1 & + & 12 x_2 & & & & \\
\text{s.t.} \quad & & 13 x_1 & + & x_2 & + & 12 x_3 & \geq 5 & \tag{constraint 1} \\
& & x_1 & & & + & x_3 & \leq 16 & \tag{constraint 2} \\
& & 15 x_1 & + & 201 x_2 & & & = 14 & \tag{constraint 3} \\
& & & & \rlap{x_i \geq 0, i = 1, 2, 3}
\end{alignat}
```

$$\begin{aligned} \max \quad & z = x_1 + 12x_2 \\ \text{s.t.} \quad & 13x_1 + x_2 + 12x_3 \geq 5 && \text{(constraint 1)} \\ & x_1 + x_3 \leq 16 && \text{(constraint 2)} \\ & 15x_1 + 201x_2 = 14 && \text{(constraint 3)} \\ & x_i \geq 0, i = 1, 2, 3 \end{aligned}$$

We treat `max`, `z`, each variable, $\pm$ sign and RHS as one separate column, while leaving an extra empty column on the right. Then we count the number of separators `&`, add one into this number then divide it by two. (e.g. $(9 + 1) \div 2 = 5$)

`\rlap` is used so that the last row spans over one column.

Optional: `\tag` is used to label the constraints.

Change MATLAB/Octave matrices to L^AT_EX code

To get fractions, execute `format rat` at the beginning.

Writing manually the L^AT_EX code for a matrix with many rows and columns in Octave is tedious. The

Octave function

```
strcat("\begin{bmatrix}\n",strrep(strrep(mat2str(A)," "," & "), ...
";"," " \\\n")(2:end-1),"\n\\end{bmatrix}\n")
```

converts

```
A = [1 2 2; 2 3 4; 4 4 2]
A =
```

```

1  2  2
2  3  4
4  4  2
```

to

```
\begin{bmatrix}
1 & 2 & 2 \\
2 & 3 & 4 \\
4 & 4 & 2
\end{bmatrix}
```

so that pasting the generated code gives

$$\begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & 4 \\ 4 & 4 & 2 \end{bmatrix}.$$

Simplex tableaux

Since the coefficient of the objective value variable z *never* changes, my habit is to omit the z -column to save ink.

Normal simplex tableau

```
\begin{array}{rrrrrr|r}
& x_1 & x_2 & s_1 & s_2 & s_3 & \\
s_1 & 0 & 1 & 1 & 0 & 0 & 8 \\
s_2 & 1 & -1 & 0 & 1 & 0 & 4 \\
s_3 & 1 & 1 & 0 & 0 & 1 & 12 \\
& -1 & -1 & 0 & 0 & 0 & 0
\end{array}
```

	x_1	x_2	s_1	s_2	s_3	
s_1	0	1	1	0	0	8
s_2	1	-1	0	1	0	4
s_3	1	1	0	0	1	12
	-1	-1	0	0	0	0

It can be stacked up to give an illustration of the entering of variables at different stages.

```
\begin{array}{rrrrrrrr|rr}
& x_1 & x_2 & s_1 & s_2 & s_3 & w & & \text{ratio} \\
s_1 & 0 & 1 & 1 & 0 & 0 & 0 & 8 & - \\
w & 1^* & -1 & 0 & -1 & 0 & 1 & 4 & 4 \\
s_3 & 1 & 1 & 0 & 0 & 1 & 0 & 12 & 12 \\
& 1 & -1 & 0 & -1 & 0 & 0 & 4 & \\
s_1 & 0 & 1 & 1 & 0 & 0 & 0 & 8 & \\
x_1 & 1 & -1 & 0 & -1 & 0 & 1 & 4 & \\
s_3 & 0 & 2 & 0 & 2 & 1 & -1 & 8 & \\
& 0 & 0 & 0 & 0 & 0 & -1 & 0 & 
\end{array}
```

```
\end{array}
```

	x_1	x_2	s_1	s_2	s_3	w	ratio	
s_1	0	1	1	0	0	0	8	—
w	1*	−1	0	−1	0	1	4	4
s_3	1	1	0	0	1	0	12	12
	1	−1	0	−1	0	0	4	
s_1	0	1	1	0	0	0	8	
x_1	1	−1	0	−1	0	1	4	
s_3	0	2	0	2	1	−1	8	
	0	0	0	0	0	−1	0	

Dual simplex tableau

```
\begin{array}{rrrrrrrr|rr}
& x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & x_7 & & \\
x_4 & 0 & -3 & 7 & 1 & 0 & 0 & 2 & 2M & -4 \\
x_5 & 0 & -9 & 0 & 0 & 1 & 0 & -1 & -M & -3 \\
x_6 & 0 & 6 & -1 & 0 & 0 & 1 & -4^* & -4M & +8 \\
x_1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & M & \\
& 0 & 1 & 1 & 0 & 0 & 0 & 2 & 2M & \\
\text{ratio} & & & 1 & & & & 1/2 & & \\
\end{array}
```

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	
x_4	0	−3	7	1	0	0	2	$2M - 4$
x_5	0	−9	0	0	1	0	−1	$-M - 3$
x_6	0	6	−1	0	0	1	−4*	$-4M + 8$
x_1	1	0	1	0	0	0	1	M
	0	1	1	0	0	0	2	$2M$
ratio			1				1/2	

It can be stacked up to give a theoretical illustration of what happens in the upcoming steps.

	x_1	x_2	x_3	s_1	s_2	s_3	
s_1	-2	0	-2	1	0	0	-60
s_2	-2	-4*	-5	0	1	0	-70
s_3	0	-3	-1	0	0	1	-27
	8	10	25	0	0	0	0
ratio	-4	-5/2	-5				
s_1	-2*	0	-2	1	0	0	-60
x_2	1/2	1	5/4	0	-1/4	0	35/2
s_3	3/2	0	11/4	0	-3/4	1	51/2
	3	0	25/2	0	5/2	0	-175
ratio	-3/2		25/4				
x_1	1	0	1	-1/2	0	0	30
x_2	0	1	3/4	1/4	-1/4	0	5/2
s_3	0	0	5/4	3/4	-3/4*	1	-39/2
	0	0	19/2	3/2	5/2	0	-265
ratio					...		
x_1	1	0	1	-1/2	0	0	30
x_2	0	1	1/3	0	0	-1/3	9
s_2	0	0	-5/3	-1	1	-4/3	26
	0	0	41/3	4	0	10/3	-330

Duality

A picture is worth [a thousand words](#) ^[1].

$$\begin{array}{ccc}
 \begin{array}{ll}
 \max & z = c^T x \\
 \text{s.t.} & Ax \leq b \\
 & x \geq 0 \\
 & (\mathcal{PC}) \\
 \text{add } \downarrow \text{slack var} \\
 \max & z = c^T x \\
 \text{s.t.} & Ax + s = b \\
 & x, s \geq 0 \\
 & (\mathcal{PS})
 \end{array}
 &
 \begin{array}{c}
 \text{duality} \\
 \Leftrightarrow \\
 \\
 \text{duality} \\
 \Leftrightarrow \\
 \text{some steps skipped}
 \end{array}
 &
 \begin{array}{ll}
 \min & v = b^T y \\
 \text{s.t.} & A^T y \geq c \\
 & y \geq 0 \\
 & (\mathcal{DC}) \\
 \text{minus } \downarrow \text{surplus var} \\
 \min & v = b^T y \\
 \text{s.t.} & A^T y - t = c \\
 & y, t \geq 0 \\
 & (\mathcal{DS})
 \end{array}
 \end{array}$$

[1] <https://math.stackexchange.com/q/2572928/290189>

(6) It must have taken *more* than a thousand words to write *that* picture though :D - **Mr Pie**
 How many words is a broken link worth? - **mmesser314**

[+20] [2017-10-17 02:02:56] **gen-ℤ ready to perish**

Units

While $\text{\LaTeX}$ has packages that format units, MathJax does not. For visual consistency, one should format units within the same string of MathJax code as the value to which it corresponds, separating the

value and unit with `\spacebackslash-space` since the [BIPM](#) ^[1] recommends a small space between the value and units. In addition, follow the below conventions for formatting values and units:

Decimal Separator & Digit Separation

Following the conventions of the English-speaking world, a period `.` should be used to separate the decimal part of a number from the integral part, not a comma `,`, as is common in some languages. This is because commas are already reserved for separating mathematical notation such as arguments of multivariate functions, elements of a set, and the coordinates of ordered tuples.

No punctuation should be used to separate multiples of three digits on either side of the decimal separator; instead, a small space rendered by `\,`, should be used on both sides of the decimal marker when the string of digits consists of more than four or five digits. For example,

- `4321.1234` `4321.1234`
- `54\,321.123\,45` `54\,321.123\,45`
- `0.56789` `0.56789`
- `0.567\,89` `0.567\,89`

If you use a decimal separator, you should include a digit on both sides of the separator, even if the digit is simply 0.

Powers of 10

It is preferable to write scientific or engineering notation like this: `4.15\times10^{n}` 4.15×10^n . The spacing around `\times` $\times$ is taken care of on its own, so there is no need to insert the spacing manually.

Nevertheless, if necessary, use an upright variant of the letter ‘E’ or ‘e’ to indicate order of magnitude, such as

- `\mathrm{E}\,6` $\mathrm{E}\,6$
- `\scriptsize{\mathrm{E}}\,\,\,\,\,\mathrm{E}\,6`
- `\mathrm{e}\,6` $\mathrm{e}\,6$

A small space on either side is perfectly fine and recommended.

Single Units

The symbol of any unit—especially SI units—should follow the form `\mathrm{u}`. (I have this command saved under the keyboard shortcut `usin` on my devices.) For example,

- `\mathrm{m}` m
- `\mathrm{kg}` kg
- `\mathrm{ft.}` $\mathrm{ft.}$

Do not use a period with symbolic units; do use a period with abbreviated units.

Units with a Dot Multiplier

Multiplied units conjoined by a dot should follow the form `\mathrm{u}\!\!\cdot\!\!\mathrm{v}` $\mathrm{u}\cdot\mathrm{v}$. (I have this sequence of commands saved under the keyboard shortcut `umul` on my devices.) Because of how `\cdot` is designed (i.e., to separate numbers), the small negative space `\!` on either side maintains uniform spacing throughout the whole compound unit. For example,

- `\mathrm{N}\!\!\cdot\!\!\mathrm{m}` $\mathrm{N}\cdot\mathrm{m}$
- `\mathrm{s}\!\!\cdot\!\!\mathrm{A}` $\mathrm{s}\cdot\mathrm{A}$

Do not use `\times` $\times$ as a separator.

Units with a Solidus Separator

Divided units conjoined by a solidus should follow the form `\left.\mathrm{u}\middle/\mathrm{v}\right.` u/v . (I have this sequence of commands saved under the keyboard shortcut `udiv` on my devices.) The extra markdown is to ensure that solidus stretches the entire height of the unit, especially when exponents are involved. For example,

- `\left.\mathrm{J}\middle/\mathrm{s}\right.` J/s
- `\left.\mathrm{m}\middle/\mathrm{s}^2\right.` m/s^2

You may include small negative spaces `\!` on either side of the solidus if you please.

Exponents

Exponents can be rendered with the standard MathJax markdown. The carat and number should immediately follow the closing brace of the `\mathrm{}` argument. For example,

- `\mathrm{m}^2` m^2
- `\left.\mathrm{m}\middle/\mathrm{s}^2\right.` m/s^2

Parentheses

Parentheses can also be rendered with standard MathJax markdown using `\left(` and `\right)` outside the argument of `\mathrm{}`. For example,

- `\left.\mathrm{kg}\!\cdot\!\mathrm{m}^2\middle/\left(\mathrm{C}\!\cdot\!\mathrm{s}\right)\right.` $kg\cdot m^2/(C\cdot s)$

Exponents in Place of Separators

If you prefer to use no separators and only powers, separator each single `\mathrm{}` with a small space `\,`, and use exponents as necessary. For example,

- `\mathrm{m}\,,\mathrm{s}^{-2}` $m s^{-2}$
- `\mathrm{s}^{-1}\,,\mathrm{mol}` $s^{-1} mol$

Examples in Context

`\mu_0=4\pi\times10^{-7} \ \ \left.\mathrm{\mathrm{T}\!\cdot\!\mathrm{m}}\middle/\mathrm{A}\right.`

$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$$

`180^\circ=\pi \ \ \mathrm{rad}`

$$180^\circ = \pi \text{ rad}$$

`N_A = 6.022\times10^{23} \ \ \mathrm{mol}^{-1}`

$$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$$

[1] <https://www.bipm.org/en/publications/si-brochure/>

(2) @SamuelMuldoon Please don't use `\mathcal{MathJax}` to write MathJax. - **The Amplitwist**

30

[+18] [2016-11-02 14:35:26] **hardmath**

Mixing code and MathJax formatting on lines

To give an example of how this might be useful, I wanted to express an algorithm in more or less the same indentation and symbolic way it appears in a paper.

On my desktop browsers (Chrome, Firefox) the following appears reasonably well spaced and indented, but loses indentation on my Android smartphone:

```

Input: positive integer  $n$ 
Output: Tangent numbers  $T_1, \dots, T_n$ 
 $T_1 \leftarrow 1$ 
for  $k$  from 2 to  $n$ 
   $T_k \leftarrow (k-1)T_{k-1}$ 
for  $k$  from 2 to  $n$ 
  for  $j$  from  $k$  to  $n$ 
     $T_j \leftarrow (j-k)T_{j-1} + (j-k+2)T_j$ 
return  $T_1, T_2, \dots, T_n$ .

```

The source can be examined for specific techniques, but the basic trick is that a MathJax dollar-delimiter can follow a closing back-tick code delimiter, but an opening back-tick should be preceded by a space when following the (closing) dollar-sign delimiter.

Here is a version using `\phantom` rather than code monospacing to produce indents and tweaking the spacing between code and MathJax expressions with `\;`, so that the results appear clear on Android browsers:

```

Input: positive integer  $n$ 
Output: Tangent numbers  $T_1, \dots, T_n$ 
 $T_1 \leftarrow 1$ 
for  $k$  from 2 to  $n$ 
   $T_k \leftarrow (k-1)T_{k-1}$ 
for  $k$  from 2 to  $n$ 
  for  $j$  from  $k$  to  $n$ 
     $T_j \leftarrow (j-k)T_{j-1} + (j-k+2)T_j$ 
return  $T_1, T_2, \dots, T_n$ .

```

(5) But this is why we have `\space`, `\quad`, and `\qquad` - **Simply Beautiful Art**

(3) While those are among the ways $\LaTeX$ provides control over spacing, they do not suffice for mixing code and MathJax formatting on a line. - **hardmath**

(2) I'm not sure if the topic of mixing other code in is well-suited here. - **Simply Beautiful Art**

(1) I would write the code in TeX using `\texttt` if I were you. Regardless, this answer probably does not belong here. - **pzp**

@pzp: Thanks, that is an interesting suggestion. - **hardmath**

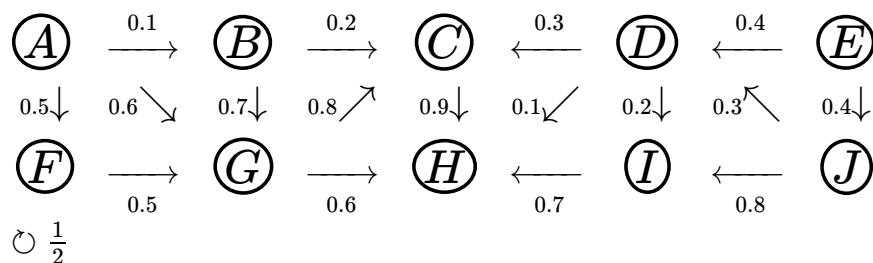
@hardmath you can shorten the `<code></code>` spacers a bit by writing `<codde/>`, at least in my Jupyter notebooks in Chrome. - **Reb.Cabin**

31

[+15] [2020-01-14 15:19:26] Jam

Markov Chains

This is a heuristic solution for coding Markov chains in Mathjax using a combination of commutative diagrams, the encircle tool and font sizes. There are a few minor issues with this method, for instance the arrows' ends should be attached closer to their targets. Also, it lacks double-headed diagonal arrows and it is difficult to attach probabilities to diagonal arrows. Therefore, it's mostly useful for small chains.



```
\require{enclose}
\begin{array}{ccccccccc}
\Large{\enclose{circle}{A}} & \xrightarrow{0.1} & \Large{\enclose{circle}{B}} & \xrightarrow{0.2} & \Large{\enclose{circle}{C}} & \xleftarrow{0.3} & \Large{\enclose{circle}{D}} & \xleftarrow{0.4} & \Large{\enclose{circle}{E}} \\
0.5\downarrow & 0.6\searrow & 0.7\downarrow & 0.8\nearrow & 0.9\downarrow & 0.1\swarrow & 0.2\downarrow & 0.3\swarrow & 0.4\downarrow \\
\Large{\enclose{circle}{F}} & \xrightarrow{0.5} & \Large{\enclose{circle}{G}} & \xrightarrow{0.6} & \Large{\enclose{circle}{H}} & \xleftarrow{0.7} & \Large{\enclose{circle}{I}} & \xleftarrow{0.8} & \Large{\enclose{circle}{J}} \\
\circlearrowright \frac{1}{2} & & & & & & & & 
\end{array}
```

Credit to [Zev Chonoles](#)^[1] for the commutative diagram.

[1] <https://math.meta.stackexchange.com/a/2325/161490>

Is there is code that generates Feynman diagrams? - **Daniel Vainshtein**

32

[+13] [2019-03-20 05:20:49] **Landon**

Tables

W/ Sub-Variables

After spending too much time searching for a way to make tables of this form (to no avail), I spent even longer searching for the pieces (of which most were found [here](#)^[1]) to Frankenstein my own. I made this table for a [combinatorics q](#)^[2] on MSE...

	Ordering	
Repetition	w/	w/o
w/	$P_r^n = n^r$	$C_r^n = \binom{n}{r} = \binom{n+r-1}{r}$
w/o	$nPr = \frac{n!}{(n-r)!}$	$nCr = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

W/o Sub-Variables

While searching, I found several tables of this form...

Day	Credit	Debit	Total
0	0	0	10000
1	100	500	9600
2	0	400	10000
3	1000	500	10500

[1] <https://math.meta.stackexchange.com/q/5020/490122>

[2] <https://math.stackexchange.com/a/3132211/490122>

I still consider myself a MathJax novice so please comment any code simplifications/corrections & I'll make the edit. - **Landon**

(2) Related: [How do I insert a table when asking a question?](#) - **Martin Sleziak**

@MartinSleziak That was 1 of the questions I came across while searching for a way to make tables w/ sub-variables. -

[+9] [2020-07-22 10:54:02] user803596

Overlaying symbols

(using negative spacing)

To overlay the $\wedge$ `\wedge` and the $\bigcirc$ `\bigcirc`, to make the Kulkarni Nomizu Product ^[1]:

$$\bigcirc\wedge$$

Which is `\mathbin{\rlap{\,\,\wedge}\bigcirc}`.

Just for another example: Overlaying of $\}$ `\}` and $\div$ `\div`:

$$\}\div$$

Which is `\rlap{\,\,\,\,}\div`.

The command `\rlap{c1}c2` prints the character `c1` with zero-width on the right-hand side of the current position, so that `c2` overlaps with `c1`. In practice, you might want to

1. choose the widest character as `c2`
2. adjust the horizontal spacing by prepending `c1` with extra horizontal space `\,`.
3. if necessary, wrap up the symbol with `\mathbin` so that MathJax treats the symbol like an operator, and the spacing around the symbol is correct.

As you can see, the number of `\!` is different, for exact overlaying of each symbol. `\!` makes the characters left and right to it move a little bit closer.

e.g, the code `ab` produces ab .

And the code `a\!b` produces ab

[1] https://en.wikipedia.org/wiki/Kulkarni%E2%80%93Nomizu_product

(1) The required horizontal spacing is device dependent. An alternative construct is `\overset{\bullet}{\smash{\cup}}`. This introduces a possibly device dependent vertical offset that cannot be adjusted, but I think(?) is always perfectly horizontally centered which is sometimes better. Using `\mathbin` one can get $A\mathbin{\odot}B$ for instance. Somehow a further nesting is needed for a `\bigcup` version: `\operatorname*{\overset{\bullet}{\smash{\bigcup}}}_1^{\infty}U_i` - Calvin Khor

[+8] [2019-09-07 20:31:26] Antoni Parellada

Tensor indices

$T^{\alpha\beta}{}_{\gamma\delta}$

$T^{\alpha\beta}{}_{\gamma\delta}$

$T^{\alpha\beta}{}_{\gamma\delta}{}^{\lambda}$

$T^{\alpha\beta}{}_{\gamma\delta}{}^{\lambda}$

So for instance, a $(2, 2)$ -tensor would act on two covectors (ω, φ) and two vectors (v, w) to produce a real number like this:

$$[T^{\alpha\beta}{}_{\gamma\delta} e_\alpha \otimes e_\beta \otimes e^\gamma \otimes e^\delta](\omega, \varphi, v, w).$$

(4) This trick of raising nothing to an exponent is also used by some to denote e.g. transposes, $\{ \}^\mathrm{t}$! A yields tA - **Calvin Khor**

35

[+8] [2020-03-23 11:37:37] **Dávid Laczkó**

Is there a way to make the Greek letters displayed upright (non-italic)?

Asked in comment. There is a way using unicode characters, for which one can search here: <http://unicode-table.com/en>. Normal use of **phi** is $\phi = \dots$, which gives $\phi = \dots$

Looking up **phi** on the above site gives a couple of results, if one uses 'Greek Capital Letter Phi' and copy it with the button below its picture, and use it like

$\Phi = \dots$, the result is $\Phi = \dots$

One might need to experiment which symbol(s) look(s) right.

Is there a tool to visually edit (prepare) the formulas with pre-defined symbols and paste here?

For some of the formulas one can use a word processor app that supports formula edition. E.g. one can use MS Word to construct the formula, or even better use the existing pre-defined ones, like I did with the Binomial theorem, then simply select it, and copy-paste here between the desired number of '\$':

$$(x + a)^n = \sum_{k=0}^n \binom{n}{k} x^k a^{n-k}$$

If it does not look right, it might still be less time to adjust the expression than starting it from scratch, or trying to draw by hand such a thing like the above.

(1) It would be nice to use an upright greek font for the circular ratio, the golden ratio, the Euler–Mascheroni constant, and the Feigenbaum constants, leaving π , ϕ , γ , δ , and α to denote variables. - **John Bentin**

36

[+8] [2022-07-28 15:49:41] **Antoni Parellada**

Evaluated at (integrals):

To get a vertical bar to the right of an expression with the limits of integration, expressions such as $\Big|_{\dots}$ result in one-size-fits-all outputs.

$\left. \dots \right|_{\dots}^{\dots}$ works well as in the made up expression below to illustrate this feature:

$$\left(3x \left(\frac{\left(\log\left(\frac{3x^2}{6} \right) \right)^{\frac{-x^2}{8}}}{3x^{1/2}} \right) \right) \Bigg|_{x=2}^{x=8}$$

(2) Thanks for the tip. I used to use $\bigg|$, but did not know you could use a period as a delimiter - **Tyma Gaidash** Upvoted. I added [an answer about this here as well](#), only to then see your answer after-the-fact. That's okay though, as my answer expounds on it a bit more and provides additional insight. - **Gabriel Staples**

37

[+8] [2022-08-27 04:20:52] Gabriel Staples

How to draw a stretched vertical bar to indicate the bounds (upper and lower limits) after taking the anti-derivative of a definite integral

I'd like to also expound upon bullet 6 in the question, about **parenthesis**. I originally put my edit into the question there, but the primary editor of the question reverted my edits. I and others need this information and a good example, so here it is:

There are also *invisible parentheses*, denoted by `.`, as in `\left.` or `\right.`. These can take the place of *any type* of parenthesis whether it be `(`, `[`, or something else. Ex: `\left.\frac{1}{2}\right\rbrace` is $\frac{1}{2}$, and `\left.\frac{1}{2}\right)` is $\frac{1}{2}$.

To stretch a vertical bar to be tall, such as to plug in upper and lower limit values into a definite integral's antiderivative, add an *invisible* vertical bar on the left with `\left.`, and a *visible* vertical bar on the right with `\right|`. For lower and upper limits of 0 and 4, respectively, the lower limit is set with `_{\;0}`, and the upper limit with `^{\;4}`, where the `\;` is a wider space to shift the numbers to the right of the vertical bar.

Example: `$$\left.\left[\frac{x}{a+b}\right]\right|_{\;0}^{\;4}$$`
produces:

$$\left[\frac{x}{a+b}\right]\Big|_0^4$$

See also:

1. This answer I just saw after writing all this: [Evaluated at \(integrals\)](#) ^[1]
2. My answer: [Tex.StackExchange.com: How to stretch a vertical bar to be tall](#) ^[2]

[1] <https://math.meta.stackexchange.com/a/35007/454133>

[2] <https://tex.stackexchange.com/a/654522/168682>

38

[+7] [2018-02-20 13:22:44] Mr Pie

Alternative Ways of Writing in $L^A T^E X$

TYPESET FONTS

As mentioned before, you can write `$\mathtt{\{ . . . \}}$` to generate fonts like **A**, **B**, **C** and etc.

You can also produce these fonts writing `$\verb| . . . |$` which generates the same fonts **A**, **B**, **C** and etc.

And concerning different “angle fonts”, `$\angle$` generates $\angle$, `$\measuredangle$` generates $\measuredangle$ and last

but not least, `\sphericalangle` generates $\sphericalangle$. Also, `\langle...\rangle` generates $\langle...\rangle$.

Concerning different “approximation fonts”, `\approx` generates $\approx$ with `\thickapprox` generating $\thickapprox$. In addition to that, `\sim` generates $\sim$ and `\thicksim` generates $\thicksim$ with `\backsim` generating $\backsim$.

For a symbol of *contradiction*, you can write `\Rightarrow\Leftarrow` to generate $\Rightarrow\Leftarrow$ or you can write `\unicode{x21af}` to generate $\z$, which is read as *Scar* (short for Harry Potter's scar, explaining why it looks like a lightning bolt).

* * *

INEQUALITY SIGNS

You can write `\lt` or `<` to generate $<$ and `\gt` or `>` to generate $>$, with `\le` or `\leq` to generate $\leq$.

You can also produce similar *less than* inequality signs with `\leqslant` to generate $\leqslant$ and `\leqq` to generate $\leqq$. The same applies for *greater than* inequality signs, for which we just rewrite the command as `\g...` instead of `\l...` which produces $\geq$, $\geqslant$ and $\geqq$.

By putting in an n, we could form commands like `\ngtr` to generate $\ngtr$ and `\nless` to generate $\nless$ as opposed to `\not>` and `\not<`.

Also, `\ngeq = \not\geq` which generates $\ngeq$ and `\nleq = \not\leq`, generating $\nleq$.

Furthermore, putting `slant` at the end of strictly the previous two commands generates $\ngtrslant$ and $\nlessslant$.

* * *

SET CONTAINMENT

You could write `\not\subseteq` to generate $\not\subseteq$ or `\not\supseteq` to generate $\not\supseteq$.

You can write `\subsetneq` to generate $\subsetneq$ and `\supsetneq` to generate $\supsetneq$.

Or, you can write `\subsetneqq` to generate $\subsetneqq$ and `\supsetneqq` to generate $\supsetneqq$.

By striking out the n in the previous commands with qq at the end, we can generate $\subseteq$ and $\supseteq$.

Instead of `\left\{... \right\}` to generate $\{...\}$, you can write `\lbrace...\rbrace` to generate the exact same thing. For sets that contain element(s) with a single number or letter, you can also write `\{\dots\}` to generate strictly $\{...\}$ with no other smaller or larger brace sizes.

As another alternative to denoting the difference of two sets A and B , you can write `\diagdown` to generate $\diagdown$ in the set expression, $A \diagdown B$. This command though is mainly used for sets A^n and B^n . There also exists `\diagup` = $\diagup$ by the way to denote the division operation as opposed to the ordinary $/$ or `\div` = $\div$.

* * *

OLD-STYLE

For old-style notation, you can write `\eqslantless` to generate $\eqslantless$ and `\eqslantgtr` to generate $\eqslantgtr$. These notations can be used to mean the same as $\leqslant$ and $\geqslant$ which is also the same as $\leq$ and $\geq$, but if used today, they commonly represent a *not much less than* or *not much greater than* inequality sign.

If you want to write that the statement, $x > y$ **and thus** $x \neq y$, *without* any words, then you can write $x \gtrvertneqq y$ to generate $x \gtrneqq y$. If, on the other hand, you want to then write the same statement for $x < y$ then you can write $x \lesvertneqq y$ to generate $x \lesneqq y$.

Suppose you have that $x \in \mathbb{R}$ but $x \neq 0$ ($\star$) for example (like in [this question](#) ^[1]), one could write it as follows: $x \in \mathbb{R} \setminus \{0\}$ with $\setminus$ to generate $\setminus$. There is an alternative way of writing ($\star$), nonetheless.

You can write $\gtrless$ to generate $\gtrless$ which means *less than and greater than*. If $x \gtrless y$ then x is equal to a number greater than y or less than y . Therefore, $x \in \mathbb{R} \setminus \{0\}$ can also be written as $x \gtrless 0$. You can also write $\lessgtr$ to generate $\lessgtr$ which essentially means the same thing. The following commands and notation is unnecessary, for their definition is obvious.

$\gtreqless$ generates $\gtreqless$ and $\lesseqgtr$ generates $\lesseqgtr$.

$\gtreqqlless$ generates $\gtreqqlless$ and $\lesseqqgtr$ generates $\lesseqqgtr$.

[1] <https://math.stackexchange.com/questions/2658670/cauchy-problem-with-real-parameter#2658682>

or $\setminus$ could be used. - **user645636**

I removed the `\backslash`, as it is different from `\setminus`. The command `\setminus` is a binary operation, which means that it has spacing designed for things like $A \setminus B$, while `\backslash` is just the symbol itself, so $A \backslash B$ is more squashed. See [here](#) for more details. - **user1729**

Is `\diagdown` really recommended for difference of sets? I'd highly recommend `\setminus` instead, both for semantic reasons and for type setting reasons. Shouldn't we adjust this recommendation? - **emacs drives me nuts**